

Conceptual framework and identification

September 15, 2026

1 Framework

1.1 Health process

- Health of individual i of (latent) type j at age $a \in \{0, 1, \dots, \bar{a}\}$ is the type's mean path plus an individual shock:

$$h_{i,a} = \bar{h}_{j,a} + u_{i,a}, \quad \bar{h}_{j,a} \equiv H_j + \sum_{a'=1}^a d_{j,a'}, \quad (1)$$

- Age a counts from a *base age* - the start of the window being modelled - not from birth: $a = 0$ is the base age and $\bar{a}$ the end of the horizon.
- $\{H_j, (d_{j,a})_a\}$ is the *deterministic* component: H_j the type's level at the base age (so $\bar{h}_{j,0} = H_j$), $d_{j,a}$ its expected health change at age a . H_j is the accumulated outcome of everything before the window - early conditions, past decline, shocks that have landed - and nothing below requires the base age to be birth.
- Types need not be coarse groups: in the limit each individual is their own type, which is the continuous case of Section 2.2, where every person has their own (H_i, d_i) .
- $u_{i,a}$ is the *stochastic* component: $\mathbb{E}[u_{i,a} | j] = 0$ at every age, but otherwise we are agnostic as to its distribution.
- At the base age: $h_{i,0} = H_j + u_{i,0}$

1.2 What stays, what is in the moment, what evolves

- A useful reading of (1): at any age, a health history carries three kinds of information -
 - what *stays* with you: your current level - the type intercept plus anything permanent that has already landed;
 - what is *in the moment*: transitory variation and measurement error, gone by the next wave;
 - how you are *evolving*: the deterministic drift $d_{j,a}$ - the only component that predicts future *changes*.
- Prevention targets the third; treatment acts on the first two as realized (the gap, however it arose).
- Note that a permanent shock, once landed, belongs to “stays”: that is why levels bundle it with the type intercept (Section 3), and why the returns will not care about that split.

- Partly persistent shocks sit between “stays” and “in the moment”: a shock is carried forward in proportion to its memory, which is what Section 3.2 reads off the data.
- Only what is carried from one wave to the next can produce covariance between waves. This single fact drives all of Section 3: the three components leave different traces on the covariance matrix, and identification is reading them apart (Section 3.2).

1.3 Deterministic versus stochastic and targeted prevention

Prevention acts on the deterministic dynamic process (the dynamics $d_{j,a}$); treatment acts on the realized distribution of h .

To target prevention one must predict. Predictability is necessary for targetability but not sufficient. Classifying by the source of the dispersion in health, and by whether policy can act on it:

Dispersion is	Actionable	Not actionable
deterministic (across types)	targetable prevention	limits of medical science
stochastic (within type)	untargeted prevention; treatment	pure noise

Reading the four cells:

- Deterministic and actionable: decline differs predictably across types, so prevention can be aimed at the types with most to gain.
- Deterministic and not actionable: the decline is predictable but nothing can be done about it.
- Stochastic and actionable: prevention still works, because everyone shares the same mean dynamics; what dies is *targeting*, not prevention. Treatment also sits here, since it acts on realized health whatever produced it.
- Stochastic and not actionable: unpredictable and untreatable variation.

Other points to note:

- Permanent vs. transitory is secondary: persistence scales the value of prevention or treatment, but whether it can be targeted depends on whether it is deterministic vs. stochastic.
- Predictability of levels $\neq$ predictability of changes: persistent stochastic realizations predict future levels, but only the deterministic component predicts future changes (and prevention targets changes).
- Types don’t need to be literally deterministic: $d_{j,a}$ is the type-conditional expected change (e.g. a type-specific hazard $\times$ shock severity in a life-course model). Prevention targets the conditional mean; within-type persistent shocks are treatable but not ex-ante targetable.
- While we can imagine interventions that change the variance without the mean (e.g. deregulating the drug market), these are secondary to interventions that shift the mean.

Figure 1: Gain from targeting prevention on dynamics is large in (a) and zero in (b)

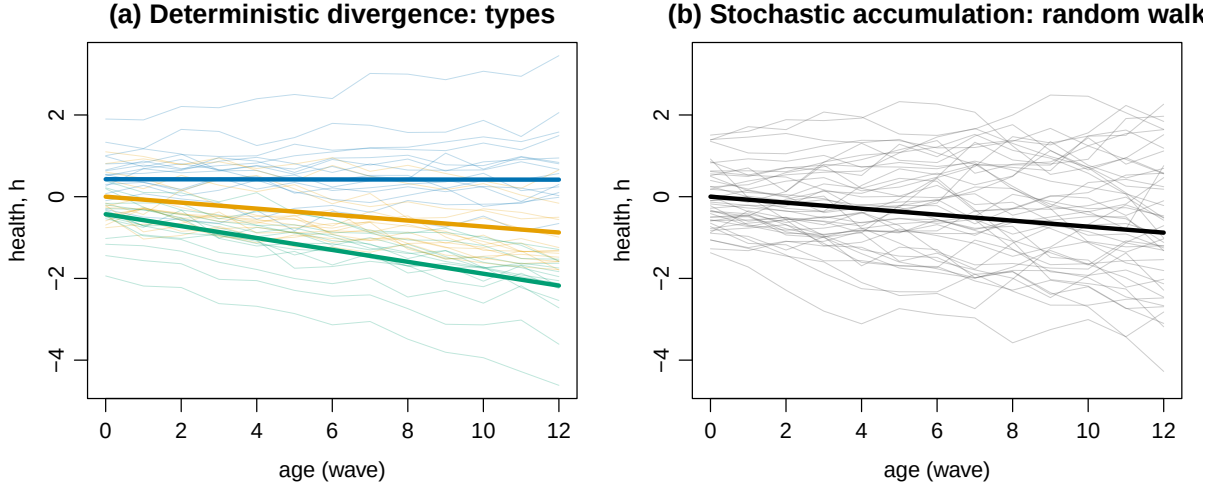


Figure 1 visualises the deterministic vs stochastic distinction: two simulated populations with the same cross-sectional health distribution (so healthcare cost is the same at every age). In panel (a) the dispersion comes from continuously distributed types (H_i, d_i) , so future decline differs predictably and targeting has value. In panel (b) it comes from a random walk, so expected future decline is the same for everyone and there is no role for targeted prevention. To work out which world we are in, we need dynamic moments.

1.4 Connection to types

Types are the object of interest because they are what makes targeted prevention possible: if decline differs predictably across groups, intensity can be aimed at the groups with most to gain - the actionable column of Section 1.3.

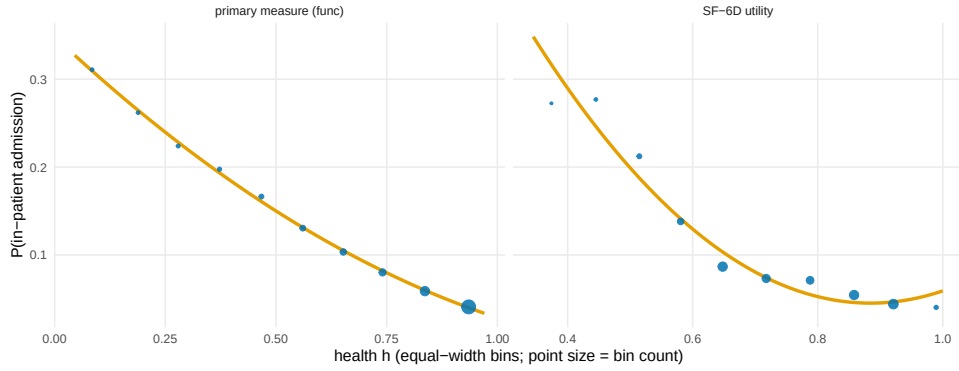
- Types represent the distribution of the deterministic components (H, d) , and clustering can estimate this. The deterministic/stochastic split maps to between vs. within variance, and to type vs. individual.
- Individuals who sit persistently above or below their type indicate that we have identified too few types: deterministic heterogeneity is leaking into the within component.
- Individuals fluctuating around their type mean, by contrast, are showing the stochastic component u .
- With the base age at birth, the three standard life-course accounts map one-to-one to the framework's second moments: *critical-period* models (early conditions leave a lasting level effect) are $\text{Var}(H)$; *accumulation* models (disadvantage builds up through the rate of decline) are $\text{Var}(d)$; *pathway* models (early disadvantage shapes the later rate) are $\text{Cov}(H, d)$.
- At a later base age, $\text{Var}(H)$ is the accumulated outcome of all three. The accounts are then separated not by one number but by how the triple $(\text{Var}(H), \text{Cov}(H, d), \text{Var}(d))$ evolves as the base age moves - which is what the data section estimates. Section 3 identifies the triple at a given base age.

2 Healthcare costs and the two instruments

2.1 Cost of poor health

- Cost of health: $c(h) = (1 - h)^2$, with the maximum level of health normalised to one.
- The quadratic is the second-order approximation to any smooth increasing convex cost, and it is what makes the first two moments of health sufficient for the policy objects below.
- Convexity is in the data: healthcare use is convex in h on every metric, with quadratic terms carrying t -statistics of 15 to 40 (Figure 2). It is mildest on the primary measure, whose cardinalisation spreads the sick finely, and strongest on the utility-weighted SF-6D - the closest thing to cost we observe.

Figure 2: In-patient admission probability against health, in equal-width bins of each measure's own units, with the fitted quadratic. Convex on both metrics: mildly on the primary measure (quadratic $t = 15$), strongly on the utility-weighted SF-6D ($t = 40$).



- Type-level expected cost at age a :

$$\mathbb{E}[c(h_{i,a}) | j] = (1 - \bar{h}_{j,a})^2 + \text{Var}(u_{i,a} | j). \quad (2)$$

- Population expected cost at age a (shares π_j , population mean health $\bar{h}_a = \sum_j \pi_j \bar{h}_{j,a}$):

$$\mathbb{E}[c(h_{i,a})] = \underbrace{(1 - \bar{h}_a)^2}_{\text{mean gap}} + \underbrace{\sum_j \pi_j (\bar{h}_{j,a} - \bar{h}_a)^2}_{\text{between types}} + \underbrace{\sum_j \pi_j \text{Var}(u_{i,a} | j)}_{\text{within types}}. \quad (3)$$

- Lifetime cost of individual i : $\mathcal{C}_i \equiv \sum_{a=0}^{\bar{a}} c(h_{i,a})$.
- Notation: moments subscripted by j ($\mathbb{E}_j$, Var_j , Cov_j , sd_j) are taken across types with weights π_j ; unsubscripted moments are across individuals. For type-level objects the two coincide (an individual inherits their type's value); the distinction matters for health itself, where $\text{Var}_j(\bar{h}_{j,a})$ is the between-type term in (3) while $\text{Var}(h_{i,a})$ is the total variance. If each individual is their own type, the between-type term collects all deterministic dispersion and the within-type term is pure noise, $\text{Var}(u_{i,a})$.

2.2 A linear case

- For intuition, and for identification later, take the type mean changes to be constant, $d_{j,a} = d_j$. Then

$$h_{i,a} = H_i + a d_i + u_{i,a}, \quad a = 0, 1, 2, \dots \quad (4)$$

a random-coefficients growth model: individual intercepts and slopes (H_i, d_i) , with types as their discrete support.

- Parameters: $V_H \equiv \text{Var}(H_i)$, $V_d \equiv \text{Var}(d_i)$, $C_{Hd} \equiv \text{Cov}(H_i, d_i)$, $v_a \equiv \text{Var}(u_{i,a})$ (written σ_u^2 when it does not vary with age). Sign: $C_{Hd} > 0 \Leftrightarrow$ types worse off at the base age decline faster; $C_{Hd} < 0 \Leftrightarrow$ catching up.
- The parameters map to observed moments as (with $\bar{H} \equiv \mathbb{E}[H_i]$, $\bar{d} \equiv \mathbb{E}[d_i]$)

$$\begin{aligned}\bar{h}_a &= \bar{H} + a\bar{d}, & \text{Var}(h_a) &= V_H + 2aC_{Hd} + a^2V_d + v_a, \\ \text{Cov}(h_s, h_t) &= V_H + (s+t)C_{Hd} + stV_d + \text{Cov}(u_s, u_t).\end{aligned}\tag{5}$$

- Section 3 shows how we can invert this map: the parameters (V_H, C_{Hd}, V_d) and the noise are recovered from observed moments.
- The linear case carries the intuition and is what Section 3 identifies. In the data, decline is not straight over long spans, and we add a quadratic age term; that is a larger deterministic basis in the general template of Appendix D ($f = \{1, a, a^2\}$), not a different framework - it costs extra parameters and hence extra waves.

2.3 Treatment: ex post, on realizations

- Treatment is an ex-post proportional reduction of the realized gap $h \mapsto h + \tau(1 - h)$, so each age- $a \geq r$ cost term scales by $(1 - \tau)^2$.

Result 1 (Return to treatment). *The marginal return to treatment beginning at age r is*

$$T(r) \equiv -\left.\frac{\partial \mathbb{E}[\mathcal{C}_i(\tau; r)]}{\partial \tau}\right|_{\tau=0} = 2 \sum_{a=r}^{\bar{a}} \mathbb{E}[(1 - h_{i,a})^2] = 2 \sum_{a=r}^{\bar{a}} \left[(1 - \bar{h}_a)^2 + \text{Var}(h_{i,a}) \right]. \tag{6}$$

- The factor of 2 is the derivative of the square: scaling the gap by $(1 - \tau)$ makes the age- a cost $(1 - \tau)^2(1 - h_{i,a})^2$. It appears in every return for the same reason, so it cancels in every comparison below.
- Treatment leaves the underlying process untouched: it rescales the realized gap at each age, and does not change the health an individual brings into the next age.
- Read plainly, $T(r)$ is twice the expected cost still to come from age r onwards.
- Treatment works through the total variance - between *and* within, including the stochastic component.
- $T(r)$ thus requires only cross-sectional moments: the treatment/healthcare-cost side of the framework is directly observable (no identification problem).

[A potential extension for the counterfactuals we could highlight here - discount future ages by survival, $\sum_{a \geq r} S(a | r)(\cdot)$. Works two ways: prevention less valuable as the sickest will die, but prevention could raise survival chances.]

2.4 Prevention: ex ante, on deterministic dynamics

- Prevention of intensity μ from age r onwards works through the deterministic change:

$$\Delta h_{i,a}(\mu) = (1 - \mu) d_{j,a} + \Delta u_{i,a} \quad (a \geq r) \quad \implies \quad h_{i,a}(\mu) = h_{i,a} + \mu(\bar{h}_{j,r-1} - \bar{h}_{j,a}), \tag{7}$$

since $-\mu \sum_{a'=r}^a d_{j,a'} = \mu(\bar{h}_{j,r-1} - \bar{h}_{j,a})$: the health gained is μ times the type's *preventable decline*, the mean health lost between the pre-intervention age $r - 1$ and age a (> 0 for declining types). The first change scaled is the one from $r - 1$ into r , so prevention takes a share μ of the deficit that would open during year r : think of it as an intervention at the start of year r , after $h_{i,r-1}$ is observed.

- Unlike treatment, prevention compounds: it scales down every future change, so the health it saves accumulates over the remaining horizon.
- Uniform prevention can be thought of as “slowing ageing”: every type's rate of change is scaled by $(1 - \mu)$, so trajectories stretch out in age, while treatment repairs the gaps that ageing has already produced.
- Prevention shifts means only here. We could of course allow interventions on the variance by letting μ also scale $\text{Var}(u_{i,a} | j)$ in (3), but that's an extension.

[An alternative set up could be flagged here and tried in the counterfactuals - instead of scaling d_j , let prevention move a person to a better type, so the return is the gap between two type paths rather than a derivative along one.]

Result 2 (Return to prevention, type/individual level).

$$P_j(r) \equiv - \left. \frac{\partial \mathbb{E}[C_i(\mu;r)|j]}{\partial \mu} \right|_{\mu=0} = 2 \sum_{a=r}^{\bar{a}} (1 - \bar{h}_{j,a}) (\bar{h}_{j,r-1} - \bar{h}_{j,a}). \quad (8)$$

- Value = gap $\times$ preventable decline, summed over remaining life, so large when the type is *already* far from full health and has a lot of decline coming up.
- Sign convention: the returns are the *signed* negative derivatives. $T(r)$ is always positive; $P_j(r)$ is positive for net-declining types and negative for improving ones (for whom scaling the dynamics down means scaling improvement down). The signed form is what aggregates exactly in Result 3.

Result 3 (Return to *uniform* prevention, population level). *For a uniform policy - the same μ applied to everyone,*

$$P(r) = \sum_j \pi_j P_j(r) = 2 \sum_{a=r}^{\bar{a}} \left[(1 - \bar{h}_a) (\bar{h}_{r-1} - \bar{h}_a) + \text{Cov}_j(1 - \bar{h}_{j,a}, \bar{h}_{j,r-1} - \bar{h}_{j,a}) \right]. \quad (9)$$

- The mean channel pairs the population gap with the population's own mean decline - both directly observable.
- Even though there is no targeting here, the between-type covariance appears anyway because the intervention is multiplicative in the type's own decline $d_{j,a}$, so types with more decline get more from the same μ .
- The covariance is positive when lower-health types are also those with more preventable decline; prevention then compresses trajectories.
- In level-covariance terms the channel is $\text{Var}_j(\bar{h}_{j,a}) - \text{Cov}_j(\bar{h}_{j,a}, \bar{h}_{j,r-1})$: how far types have fanned out by age a , over and above the spread they already had when the policy started. If types kept the same spacing after $r - 1$ the two terms would be equal and the channel would vanish.

- Those two moments are the diagonal cell at a and the $(r-1, a)$ cell of the covariance matrix, with the noise stripped out. This already introduces our framework that prevention returns are priced by particular cells of the covariance surface, and Section 3 is about recovering them.
- The gain from *deliberately* targeting (choosing μ_j by type) is measured against this uniform benchmark in Result 7.
- Decomposition of the between-type channel: split the health gap into initial gap, pre-intervention decline, and preventable decline, $1 - \bar{h}_{j,a} = (1 - H_j) + (H_j - \bar{h}_{j,r-1}) + (\bar{h}_{j,r-1} - \bar{h}_{j,a})$. Then:

$$\begin{aligned} & \text{Cov}_j(1 - \bar{h}_{j,a}, \bar{h}_{j,r-1} - \bar{h}_{j,a}) = \\ & \underbrace{\text{Cov}_j(1 - H_j, \bar{h}_{j,r-1} - \bar{h}_{j,a})}_{\text{initial disadvantage} \times \text{later decline}} + \underbrace{\text{Cov}_j(H_j - \bar{h}_{j,r-1}, \bar{h}_{j,r-1} - \bar{h}_{j,a})}_{\text{persistence of decline}} + \underbrace{\text{Var}_j(\bar{h}_{j,r-1} - \bar{h}_{j,a})}_{\text{dispersion of decline}}. \end{aligned} \quad (10)$$

The three channels are the pathway margin (initial disadvantage $\times$ later decline), the persistence margin, and the dispersion-of-decline (accumulation) margin; each maps to moments identified in Section 3.

2.5 Comparing the two instruments

Result 4 (Prevention vs. treatment). *Compare compositions:*

$$T(r) = 2 \sum_{a \geq r} \left[(1 - \bar{h}_a)^2 + \underbrace{\text{Var}_j(\bar{h}_{j,a})}_{\text{between}} + \underbrace{\sum_j \pi_j \text{Var}(u_{i,a} | j)}_{\text{within}} \right], \quad (11)$$

$$P(r) = 2 \sum_{a \geq r} \left[(1 - \bar{h}_a)(\bar{h}_{r-1} - \bar{h}_a) + \text{Cov}_j(1 - \bar{h}_{j,a}, \bar{h}_{j,r-1} - \bar{h}_{j,a}) \right]. \quad (12)$$

Treatment reaches all realized dispersion, including the stochastic within-type part, but only scales down gaps that have already opened. Prevention averts future gaps but must be assigned ex ante on predictions. A uniform policy pays the population-average preventable decline $\bar{h}_{r-1} - \bar{h}_a$: intensity goes to slow decliners and improvers as well as to steep decliners, and the same total intensity concentrated on the steep decliners would buy more. Its between-type channel is bounded by deterministic heterogeneity.

- Both returns are per-person objects: they are derivatives of the expected lifetime cost of a randomly drawn individual, so an economy-wide total is N times either one and no comparison here depends on population size.
- A caveat on the treatment side: if part of measured variance is measurement error rather than genuine health variation, it should be excluded from $\text{Var}(h_a)$ in $T(r)$ - the only decomposition treatment ever needs. Measurement error appears as a diagonal spike, a term that lives only on the variances (Appendix D); the prevention return never contains it. Case 4 of Section 3.3 separates the spike from genuine transitory health, which is what makes the exclusion operational.
- The value of deliberately targeting prevention is a per-person statement in a different sense: returns differ across people, so it matters where a unit of intensity goes. Section 2.6 prices this - what any allocation collects (Result 5), what initial health collects (Result 6), and the ceiling set by the type itself (Result 7).
- Derivations for this section are in Appendix A.

2.6 Targeting

Prevention must be assigned before outcomes are seen, so the question is not only how much intensity to use but where it goes. This subsection asks what targeting is worth: first as an identity that holds for any signal, then for the simplest feasible signal (initial health), then for the best possible signal (the type itself).

Result 5 (What targeting collects). *Let $\mu(z)$ be any allocation of prevention intensity based on information z , holding average intensity fixed. The saving over the uniform policy is*

$$\text{Cov}(\mu(z), \mathbb{E}[P_i | z]). \quad (13)$$

Moving one unit of intensity from a person with signal z' to a person with signal z saves $\mathbb{E}[P_i | z] - \mathbb{E}[P_i | z']$.

- Targeting is worth exactly the covariance between where intensity goes and what it buys there.
- It is zero when predicted returns do not vary with z : a signal that does not separate people is worth nothing, however well it predicts health.
- The per-person form is the practical one - what matters is the difference in predicted returns between the person treated and the person not treated.
- The value of a signal is therefore how much predicted returns differ across people who differ in it. The rest of this subsection computes that for two signals.
- Per unit of dispersion in intensity, no rule based on z can save more than $\text{sd}(\mathbb{E}[P_i | z])$ (Appendix B).

Result 6 (Targeting on initial health). *Let the signal be initial health $h_{i,0}$ - the level at the base age. Since P_i depends on the individual only through their type,*

$$\text{Cov}(h_{i,0}, P_i(r)) = \text{Cov}_j(\bar{h}_{j,0}, P_j(r)), \quad (14)$$

and the linear projection of $P_i(r)$ on $h_{i,0}$ has slope

$$\beta(r) = \frac{\text{Cov}(h_{i,0}, P_i(r))}{\text{Var}_j(\bar{h}_{j,0}) + v_0}. \quad (15)$$

If the type mean paths have zero third central moments (for example jointly normal type parameters), the covariance splits by age into two channels:

$$\text{Cov}(h_{i,0}, P_i(r)) = 2 \sum_{a \geq r} \left[\underbrace{-(\bar{h}_{r-1} - \bar{h}_a) \text{Cov}_j(\bar{h}_{j,0}, \bar{h}_{j,a})}_{\text{gap channel}} + \underbrace{(1 - \bar{h}_a) \text{Cov}_j(\bar{h}_{j,0}, \bar{h}_{j,r-1} - \bar{h}_{j,a})}_{\text{compounding channel}} \right]. \quad (16)$$

In the linear case $\text{Cov}_j(\bar{h}_{j,0}, \bar{h}_{j,a}) = V_H + aC_{Hd}$, and the preventable decline is exposure time $(a - r + 1)$ times the type's own slope, so

$$\text{Cov}(h_{i,0}, P_i(r)) = 2 \sum_{a \geq r} (a - r + 1) \left[\bar{d}(V_H + aC_{Hd}) + C_{Hd}(\bar{H} - 1 + a\bar{d}) \right], \quad (17)$$

the first term inside the bracket being the gap channel and the second the compounding channel.

- The expected return falls by $|\beta(r)|$ per unit of initial health.
- The gap channel is the mean preventable decline times how strongly initial health predicts *future levels*: low initial health predicts a larger future gap, and a given decline is worth more to prevent when the gap is large.
- The compounding channel is the mean gap times how strongly initial health predicts the *decline itself*. It is zero exactly when initial health does not predict subsequent decline; the gap channel survives even then.
- Under decline and compounding ($\bar{d} < 0$, $\bar{H} < 1$, $C_{Hd} \geq 0$) both channels are negative, so $\beta(r) < 0$: lower h_0 means a higher expected return.
- Both inputs are cells of the first row of the covariance matrix with the noise removed - $\text{Cov}_j(\bar{h}_{j,0}, \bar{h}_{j,a})$, and its difference from $\text{Cov}_j(\bar{h}_{j,0}, \bar{h}_{j,r-1})$. Targeting on initial health is priced by the first row and the mean path.
- h_0 is a noisy signal of the type's level: the slope is attenuated by $\lambda_0 = \text{Var}_j(\bar{h}_{j,0})/(\text{Var}_j(\bar{h}_{j,0}) + v_0)$ relative to observing that level, and the per-sd comparison by $\sqrt{\lambda_0}$. Targeting benefits from the level/noise split.
- Every input is identified in Section 3.
- The same logic at $z = h_{i,r-1}$ is Result B.2 in Appendix B.
- Proof in Appendix B.

Result 7 (Type targeting, and the ceiling). *With the type as the signal, moving one unit of intensity from type j' to type j saves $P_j(r) - P_{j'}(r)$, and treating the highest-return type rather than a random person saves*

$$\max_j P_j(r) - \sum_j \pi_j P_j(r). \quad (18)$$

No signal does better: since P_i depends on the individual only through their type, any other signal predicts returns only through what it reveals about the type, so $\text{Var}(\mathbb{E}[P_i | z]) \leq \text{Var}_j(P_j(r))$ for every z .

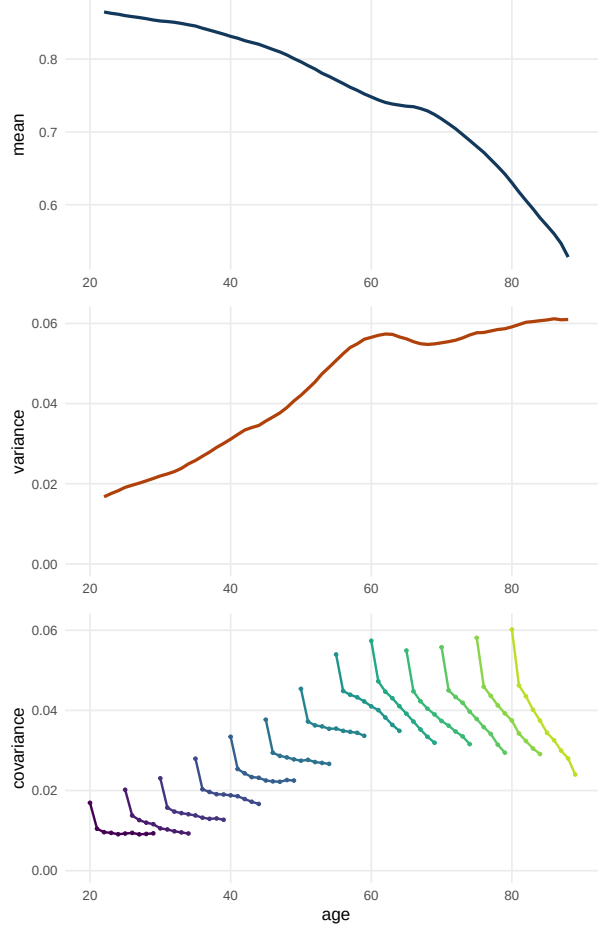
- The type is the ceiling because it is the whole of what determines the return; every other signal is a coarser view of it.
- The gap between what initial health collects (Result 6) and this ceiling is what estimating types is worth.

3 Identification

The treatment/healthcare side (Result 1) needs only cross-sectional $\text{Var}(h_a)$. The prevention side (Results 2-4) needs the between-type moments, which are not directly observable.

Figure 3 shows the raw material: the primary measure's mean path, its variance profile, and the rows of its covariance matrix. Three patterns stand out before any model. The mean falls in an accelerating way. The variance rises steeply and then, from about age 60, stops growing. And every row drops sharply in its first year - health not carried even one year - after which the young rows run flat while the old rows keep falling. The rest of the section builds the machinery that reads these patterns, and the close of the section returns to them.

Figure 3: The primary measure over the life course: the mean path, the variance profile, and the rows of the covariance matrix. The variance and covariance panels share a vertical scale, so the drop from the profile to each row leaving it is the one-year spike. Profiles are five-year centred rolling means.



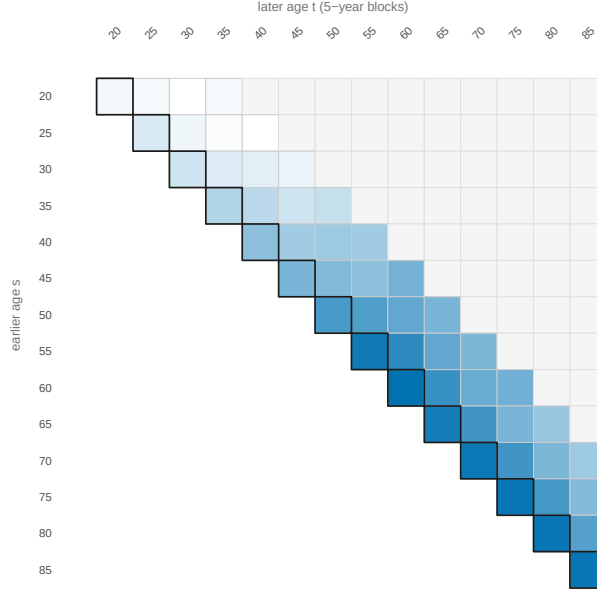
3.1 From observed moments to deterministic heterogeneity

- Every observed level covariance splits into a between-type part and a noise part:

$$\text{Cov}(h_s, h_t) = \text{Cov}_j(\bar{h}_{j,s}, \bar{h}_{j,t}) + \text{Cov}(u_{i,s}, u_{i,t}). \quad (19)$$

- The returns need exactly the between-type part: $P(r)$ is built from the between-type variances $\text{Var}_j(\bar{h}_{j,a})$ and covariances $\text{Cov}_j(\bar{h}_{j,a}, \bar{h}_{j,r-1})$, while observation delivers the sum. Identification therefore means choosing cells where the noise term is zero or has a known shape.
- In the linear case the between-type part is $V_H + (s+t)C_{Hd} + stV_d$ (Section 2.2); the proposition only requires linearity over the waves used (A1).
- Figure 4 is the object itself, in the data: every cell is one equation of the inversion.

Figure 4: The covariance surface of the primary measure, in five-year blocks. The diagonal is outlined; cells more than fourteen years off the diagonal are empty by construction - the panel is fifteen waves.



The variances alone - the diagonal of the covariance matrix - are the objects that enter costs, but they are not enough, for two reasons.

1. If the noise variances are unrestricted, the diagonal identifies nothing at all: pure noise reproduces any variance profile exactly.
 2. When the noise variance is stationary, the diagonal still cannot separate the type level V_H from the noise: they enter every diagonal cell only as the bundle $V_H + \sigma_u^2$.
- Figure 1 illustrates this: the two worlds share every cross-section, and so every diagonal cell, and differ only off the diagonal.
 - Restoring identification needs a restriction on how the noise variance moves with age, and at least one off-diagonal cell. Lemma A.1 in Appendix A states both precisely.

Assumptions.

- A1. (Window linearity) $d_{j,a} = d_j$ over the waves used.
- A2. (Orthogonality) $\mathbb{E}[u_{i,a} | H_i, d_i] = 0$ for all a . (allows type-specific shock variances.)
- A3. (Noise distribution) Case dependent.

3.2 The forces behind identification

Identification rests on a small number of forces, and each lives in a different region of the covariance matrix. The starting point is the rule of Section 1.2: only what is carried from one

wave to the next can produce covariance between waves. The deterministic part is carried for ever. Noise is carried according to its memory. Momentary noise is not carried at all.¹

The diagonal

- Every component adds variance, so every component appears in $V(a)$. On its own the diagonal separates nothing, unless the stochastic component has stationary variance.
- The force that makes it useful is stationarity: if the noise contributes the same variance at every age, differences of $V(a)$ remove it completely.
- What survives differencing is the deterministic part, because it grows with age in a known way: C_{Hd} adds a linear term and V_d a quadratic one, so the slope and curvature of the diagonal are C_{Hd} and V_d .
- This force ignores memory: any noise with constant variance drops out of the differences, so persistence never touches what the diagonal delivers.
- The force fails when v_a moves with age: on the diagonal alone, a drifting noise variance cannot be told apart from deterministic growth.

The first row

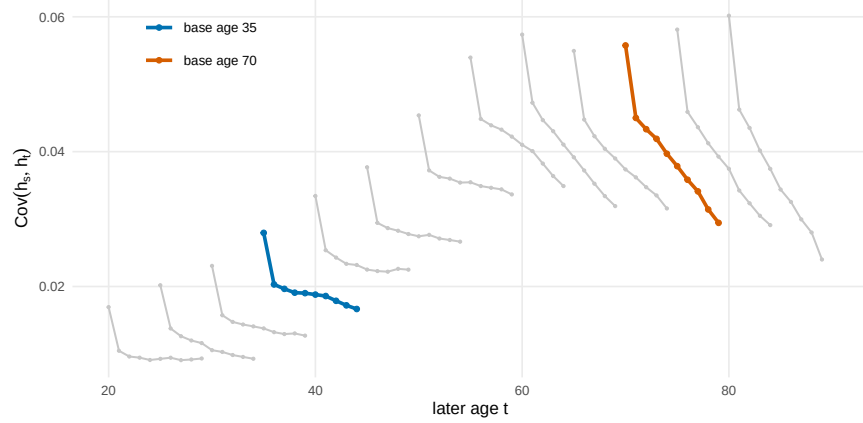
- The force here is separation by shape. Fix wave 0 and move the later wave: the deterministic part contributes a straight line, $V_H + aC_{Hd}$, because H and d are carried unchanged.
- Carried noise contributes a shape of its own, and the shape is what identifies the process. Under AR(1) it is a geometric decay $\rho^a v_0$, a share ρ surviving each step; under an MA(q) it dies after q steps; under a random walk it is flat.
- A line and a geometric decay cannot imitate each other, which is what splits the level from the noise and measures the memory. The exception is the flat case, and that is the limit below.
- A component carried to no other cell at all - a pure one-period spike - is invisible to the row entirely; the gap between the diagonal and the row's back-extrapolation is what measures it.
- The anchor $V(0)$ is part of the row: it is the $a = 0$ point of the same function. That is why the row's second differences can remove the line.
- The force has a limit: as $\rho \rightarrow 1$ the decay flattens into a constant and merges with V_H . This is the bundle - a permanent shock, once landed, is carried exactly like a type level ("what stays", Section 1.2).
- The row never contains V_d : the curvature term is stV_d , and on the first row $s = 0$. Whatever the row cannot deliver has to come from the interior.

Figure 5 shows this force in the data: the rows of the primary measure's matrix, leaving the diagonal at five-year base ages. The row from base age 35 drops off the diagonal - the spike dying - and then runs essentially flat: a line with almost no slope. The row from base age 70 drops and then keeps falling. One stationary noise process cannot produce both, because the

¹In derivative language, writing $C(s, t) = \text{Cov}(h_s, h_t)$: the derivative along the diagonal (the slope of the variance profile) mixes the drift in the noise variance with the deterministic terms, and is clean only when the noise variance is stationary; the partial derivative in t at fixed s is a row read, equal to $C_{Hd} + rV_d$ wherever the noise is absent or has died; and the cross derivative $\partial^2 C / \partial s \partial t$ - discretely the double difference, algebraically $\text{Cov}(\Delta h_s, \Delta h_t)$ - isolates the curvature V_d , because the tilt cancels.

same fade would have to appear at every base age. Either the noise moves with age, or the line's own slope has changed sign - which is the local reading of Section 3.4.

Figure 5: Rows of the covariance matrix (primary measure), each starting on the diagonal at its own base age and running nine years. The initial drop is the one-year spike; what remains is the line plus any carried noise.



The interior

- V_d multiplies st , so it appears on the diagonal (as $a^2 V_d$) and in the interior (as $st V_d$), but never in the first row.
- When the diagonal is readable, its curvature gives V_d and the interior is spare. When it is not, the interior is the only place left.
- With the row solved, two interior cells contain exactly two unknowns, V_d and v_1 : the noise they carry, $\rho^{t-s} v_s$, uses the memory the row has already measured.
- Every further cell is spare capacity: an overidentifying test, or the budget for relaxing an assumption - which is what the cases below spend it on.

Table 1: The three regions and their forces

Region	Force	Delivers	Fails when
Diagonal $V(a)$	stationary noise cancels in differences	C_{Hd}, V_d	v_a moves with age
First row $C_0(a)$	a line cannot mimic a geometric decay	level split; ρ	$\rho \rightarrow 1$: noise becomes permanent and merges with the level
Interior	$st V_d$ needs both indices to move	V_d when the diagonal cannot be read; tests otherwise	-

- The policy mapping follows the regions: the diagonal prices the uniform returns (Section 3.4), the row provides the split that targeting needs (Result 6), and the interior completes the general case and supplies the tests.

3.3 Four cases: the two curves and the interior

Two functions of the data organise everything below:

$$V(a) \equiv \text{Var}(h_a), \quad C_0(a) \equiv \text{Cov}(h_0, h_a), \quad \text{with } C_0(0) = V(0). \quad (20)$$

$V(a)$ is the diagonal of the covariance matrix and $C_0(a)$ is its first row. The interior cells $\text{Cov}(h_s, h_t)$ with $s \geq 1$ are the third region: they are where V_d has to come from once the diagonal can no longer be read, and elsewhere they are overidentifying checks. Section 3.2 explains what each region can do; the formulas that recover a case's parameters we call the *reads* of that case. Proofs are in Appendix A.

Proposition 1 (Identification within the AR-plus-spike class). *Assume A1-A2 and let the noise satisfy $\text{Cov}(u_s, u_t) = \rho^{t-s}v_s + \mathbf{1}\{s = t\}\sigma_{\varepsilon,s}^2$ for $s \leq t$, with $\rho \in [0, 1]$: a carried component with memory ρ and variances v_s , plus a one-period spike $\sigma_{\varepsilon,s}^2$. Then:*

- (i) *five waves identify V_H , C_{Hd} , V_d , ρ and both noise variances at the waves used (the last wave's two components only in sum);*
- (ii) *with the spike absent, four waves suffice (Case 3); with both noise variances stationary, four waves suffice (Case 4);*
- (iii) *with the spike absent and the carried variance stationary, three waves suffice, whatever the memory ρ (Cases 1-2);*
- (iv) *if in addition $C_{Hd} = 0$ and the noise is memoryless, two waves suffice.*

The explicit reads are given by the four cases below.

- Some restriction on the noise is necessary (here: a carried component plus a one-period spike). Identification needs at most as many parameters as there are moments, and an unrestricted noise covariance already uses all of them: with T waves there are $T(T+1)/2$ covariance moments and exactly that many free noise parameters.
- Structure on how the noise is carried - here one memory parameter and one variance per wave - is what leaves room for the deterministic block. Lemma A.1 makes the same point on the diagonal.
- Part (iii) is the substantive one: with a stationary carried variance and no spike, memory costs nothing in waves. What raises the requirement to four is not persistence but either a carried variance that moves with age (Case 3) or a component the row cannot see at all (Case 4).

Case 1: i.i.d. noise, stationary variance σ_u^2 (three waves). The two curves are

$$V(a) = V_H + 2aC_{Hd} + a^2V_d + \sigma_u^2, \quad C_0(a) = V_H + aC_{Hd} \quad (a \geq 1). \quad (21)$$

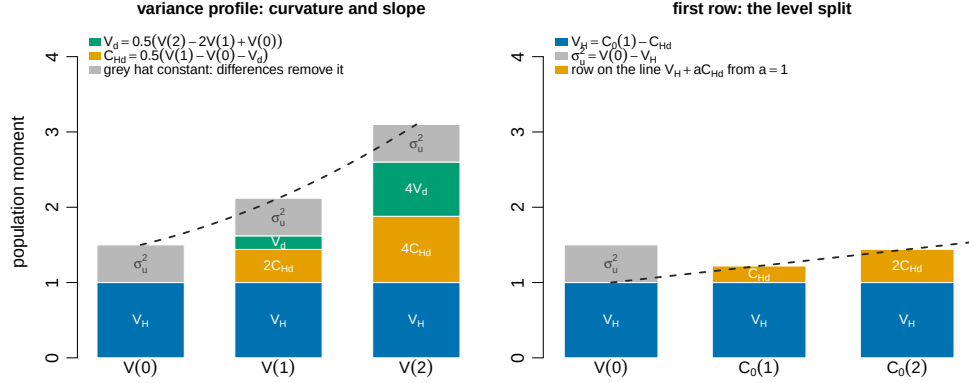
The curvature and slope of the diagonal give the evolution parameters,

$$V_d = \frac{1}{2}[V(2) - 2V(1) + V(0)], \quad C_{Hd} = \frac{1}{2}[V(1) - V(0) - V_d], \quad (22)$$

and one row cell splits the level from the noise: $V_H = C_0(1) - C_{Hd}$, $\sigma_u^2 = V(0) - V_H$.

Figure 6: Illustration of Case 1: the two curves

Case 1 (i.i.d., stationary): the profile gives C_{Hd} and V_d ; one row cell splits V_H from σ_u^2



- The noise adds the same amount to every variance, so it cancels in the differences that give C_{Hd} and V_d (Figure 6).
- Benchmark: imposing $C_{Hd} = 0$ reduces this to two waves, with $V_d = V(1) - V(0)$ - part (iii) of the proposition.

Case 2: AR(1) noise, stationary variance (three waves). Let $u_{i,a} = \rho u_{i,a-1} + \eta_{i,a}$ with $\rho \in [0, 1)$ and $u_{i,0}$ drawn from the stationary distribution. Then $V(a)$ is as in Case 1 and

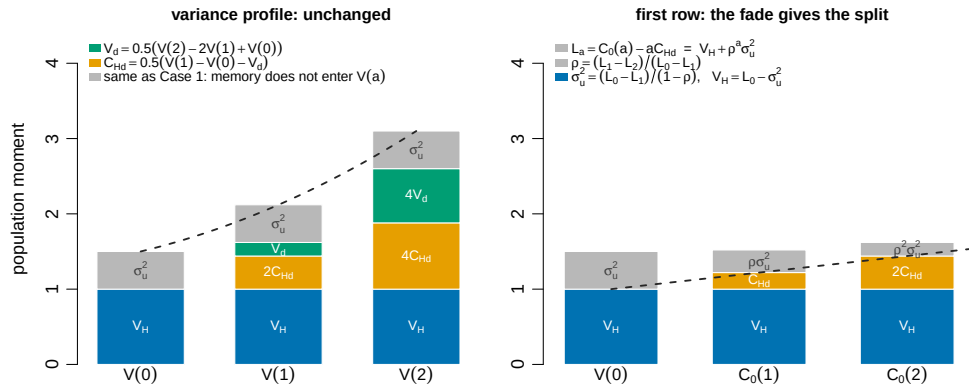
$$C_0(a) = V_H + aC_{Hd} + \rho^a \sigma_u^2. \quad (23)$$

The diagonal identification (22) is unchanged, so the tilt is already known; remove it from the row. The *detilted row* $\tilde{C}_0(a) \equiv C_0(a) - aC_{Hd}$ equals $V_H + \rho^a \sigma_u^2$, and its successive drops shrink by a factor ρ at each step:

$$\rho = \frac{\tilde{C}_0(1) - \tilde{C}_0(2)}{\tilde{C}_0(0) - \tilde{C}_0(1)}, \quad \sigma_u^2 = \frac{\tilde{C}_0(0) - \tilde{C}_0(1)}{1 - \rho}, \quad V_H = \tilde{C}_0(0) - \sigma_u^2. \quad (24)$$

Figure 7: Illustration of Case 2: the two curves

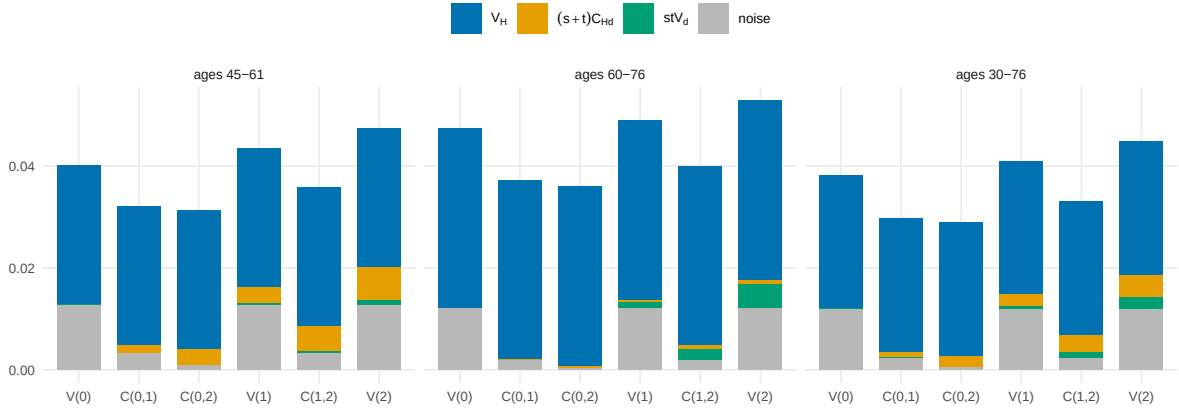
Case 2 (AR(1), stationary): the profile reads are unchanged; the row decays at ρ per step



- The diagonal identifications are identical to Case 1: noise memory never enters $V(a)$ while its variance is stationary.
- The row cells sit above the line $V_H + aC_{Hd}$ by $\rho^a \sigma_u^2$; two shrinkage steps give ρ (Figure 7).

The same read on the data: three ages three years apart, pooled over base ages within a window, with ρ profiled and the rest solved by least squares. Figure 8 draws the fitted decomposition of all six moments in every window where the answer is a valid covariance matrix.

Figure 8: Case 2 on the data (primary measure): three ages three years apart, the six moments and their fitted decomposition, in the windows where the deterministic block is a valid covariance matrix. The young window (25-41) fails and is not drawn.



- In the mid-life window (45-61) the read delivers the full parameter set: $\rho = 0.63$, a level/noise split of $V_H = 0.027$ against $\sigma_u^2 = 0.013$, and visible evolution terms with $\text{Corr}(H, d) = 0.64$.
- The window matters: at 60-76 the evolution terms nearly vanish (the diagonal is flat there), and in the young window (25-41) the read fails - V_d comes back negative, which is why it is not drawn.
- All fits match their six cells to within a third of a percent; the sign and size of the small C_{Hd} are the fragile part, taken up below and in Appendix E.

Case 3: AR(1) noise, age-specific variances v_a (four waves). The noise layer is

$$\text{Cov}(u_s, u_t) = \rho^{t-s} v_s \quad (s \leq t), \quad (25)$$

so

$$V(a) = V_H + 2aC_{Hd} + a^2V_d + v_a, \quad C_0(a) = V_H + aC_{Hd} + \rho^a v_0. \quad (26)$$

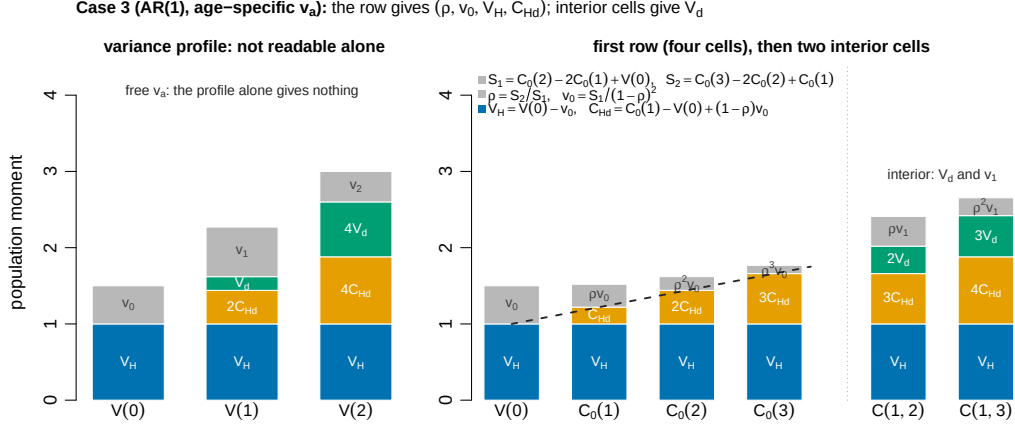
The diagonal now identifies nothing (Lemma A.1(i)), so the row must find both the line and the decay: four cells, four parameters. Second differences of the row remove the line and leave only the decay, so their ratio is the memory,

$$\rho = \frac{C_0(3) - 2C_0(2) + C_0(1)}{C_0(2) - 2C_0(1) + C_0(0)}, \quad v_0 = \frac{C_0(2) - 2C_0(1) + C_0(0)}{(1 - \rho)^2}, \quad (27)$$

and then $V_H = V(0) - v_0$ and $C_{Hd} = C_0(1) - C_0(0) + (1 - \rho)v_0$. Two interior cells give V_d : with (V_H, C_{Hd}, ρ) known, $\text{Cov}(h_1, h_2)$ and $\text{Cov}(h_1, h_3)$ leave exactly the two unknowns

436 (V_d, v_1) (closed form in Appendix A, Case 3). The remaining variances follow from the diagonal,
 437 $v_a = V(a) - V_H - 2aC_{Hd} - a^2V_d$.

Figure 9: Illustration of Case 3: the two curves, then the interior



- 438 • In Case 2 the diagonal supplied the line and the row had only to find the fade. Here the
 439 diagonal supplies nothing, so the row must find the line and the fade at once, hence the
 440 second differences rather than first (Figure 9).
- 441 • Case 3 nests Cases 1 and 2: $\rho = 0$ with free v_a is the denoising case (Appendix D); constant
 442 v_a is Case 2; $\rho = 1$ is the random walk.
- 443 • If v_a is linear in a , the noise is a random walk with constant innovation variances; the $\rho \rightarrow 1$
 444 limit is discussed in Section 3.2 and treated in full in Proposition D.1.

445 **Case 4: AR(1) noise plus one-period measurement error (four waves).** Split the
 446 shock into a carried part and a momentary part, $u_{i,a} = w_{i,a} + \varepsilon_{i,a}$: w is AR(1) with memory ρ
 447 and stationary variance σ_w^2 , and ε is measurement error - i.i.d. with variance σ_ε^2 , shared with no
 448 other cell. Then

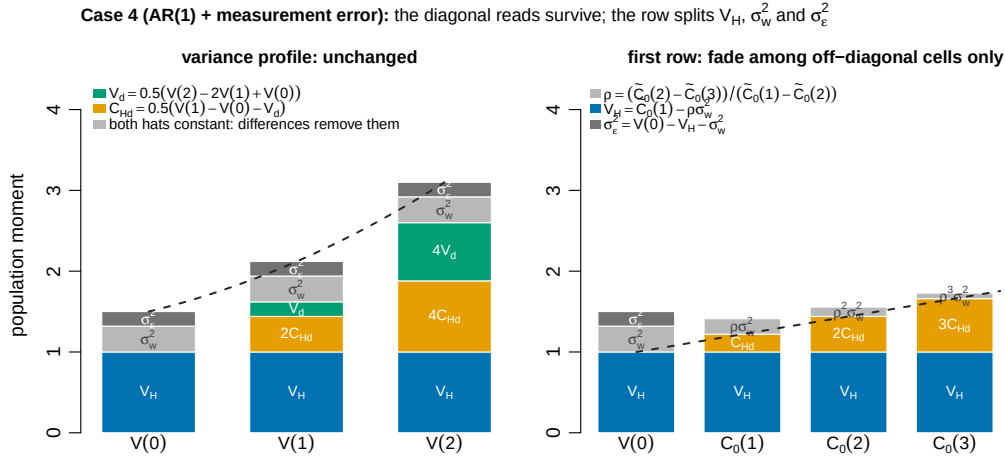
$$449 \quad V(a) = V_H + 2aC_{Hd} + a^2V_d + \sigma_w^2 + \sigma_\varepsilon^2, \quad C_0(a) = V_H + aC_{Hd} + \rho^a\sigma_w^2 \quad (a \geq 1). \quad (28)$$

450 The total noise variance is constant, so the diagonal reads (22) survive unchanged. The spike
 451 appears in every diagonal cell and in no off-diagonal cell, so the row is blind to it - and Case
 452 2's fade read breaks, because its first drop $\tilde{C}_0(0) - \tilde{C}_0(1)$ is inflated by σ_ε^2 . The repair uses only
 453 drops the spike cannot touch:

$$454 \quad \rho = \frac{\tilde{C}_0(2) - \tilde{C}_0(3)}{\tilde{C}_0(1) - \tilde{C}_0(2)}, \quad \sigma_w^2 = \frac{\tilde{C}_0(1) - \tilde{C}_0(2)}{\rho(1 - \rho)}, \quad V_H = \tilde{C}_0(1) - \rho\sigma_w^2, \quad (29)$$

455 and the spike is the anchor's excess over the fade, $\sigma_\varepsilon^2 = V(0) - V_H - \sigma_w^2$. Needing $\tilde{C}_0(3)$ is what
 456 makes this a four-wave case.

Figure 10: Illustration of Case 4: the fade is read from off-diagonal drops only; the spike lives on the diagonal alone



- The split needs memory: at $\rho = 0$ both components die in one period and only their sum is identified. Measurement error is separable from genuine transitory health only because the genuine part is carried.
- Three waves are not enough despite the parameter count (six moments, six parameters): the interior cell is implied by the diagonal differences, so the map is rank-deficient - the fourth wave is genuinely needed.
- This is the case that makes the treatment-side caveat of Section 2.5 operational: σ_ε^2 is exactly the object excluded from $\text{Var}(h_a)$ in $T(r)$, while σ_w^2 stays in.
- Sibling to Case 3: each spends the fourth wave on one relaxation - a drifting carried variance there, the spike here. Both at once is the five-wave general member of the proposition.
- Nesting: $\sigma_\varepsilon^2 = 0$ is Case 2; $\rho \rightarrow 1$ is the random walk plus measurement noise of Proposition D.1.
- On the data the case is admissible where Case 2 is (from age 60): there the spike is a sixth to a fifth of the variance and the carried part is slow, $\rho \approx 0.9$ (full record in the companion `testing.metrics.tex`). The estimation-side counterpart is the bundle fit of Appendix E: with ρ near one, the carried part is folded into the level rather than estimated.

Estimation and checks.

- The ratios (24) and (27) divide small differences: exact in population, noisy in samples.
- In estimation the parameters would be chosen to fit all cells jointly (minimum distance, as in the earnings-dynamics literature); the closed forms above are the just-identified special cases that use exactly as many cells as parameters.
- Minimum distance works because the map is invertible, which is what the proposition establishes, and the cases say which cells carry which parameter - hence what drives precision.
- Every cell a case does not use is an overidentifying check.

Linear-regression representation. Every case can equivalently be expressed through individual growth regressions: OLS of $h_{i,a}$ on $(1, a)$ person by person, keeping intercepts, slopes

and residuals. The coefficient moments are an invertible linear transform of the level moments, and the transform depends only on which waves are used, not on the noise, so the two representations always carry the same information. What is case-specific is the correction for noise, not the representation (Appendix C).

3.4 Back to the returns

In the linear case of Section 2.2: $h_{i,a} = H_i + a d_i + u_{i,a}$, the type mean paths are straight lines, $\bar{h}_{j,a} = H_j + a d_j$, and Proposition 1 has just recovered the parameters (V_H, V_d, C_{Hd} and the noise parameters) of (5) from the covariance matrix. Substituting back into the returns of Section 2:

- The preventable decline becomes window length times the type's own slope, $\bar{h}_{j,r-1} - \bar{h}_{j,a} = -(a - r + 1) d_j$, so the between-type channel of (9) reduces to identified objects,

$$\text{Cov}_j(1 - \bar{h}_{j,a}, \bar{h}_{j,r-1} - \bar{h}_{j,a}) = (a - r + 1)(C_{Hd} + aV_d), \quad (30)$$

and the mean path is $\bar{h}_a = \bar{H} + a\bar{d}$, so

$$P(r) = 2 \sum_{a=r}^{\bar{a}} \left[(1 - \bar{h}_a)(\bar{h}_{r-1} - \bar{h}_a) + (a - r + 1)(C_{Hd} + aV_d) \right]. \quad (31)$$

Compounding ($C_{Hd} > 0$) raises the return, but even with $C_{Hd} = 0$ the term aV_d is positive. Summing the second term over a gives the single combination $K(r) = A(r)C_{Hd} + B(r)V_d$ that a fixed starting age needs.²

- The three channels of (10) are now visible in the parameters: the pathway margin is $(a - r + 1)C_{Hd}$, the persistence margin $(r - 1)(a - r + 1)V_d$, and the accumulation margin $(a - r + 1)^2 V_d$.
- The two V_d pieces sum to $(a - r + 1)a V_d$, so the life-course decomposition and (31) are the same object; a later start puts more of the V_d term into persistence and less into accumulation.
- The mix shifts with r as each sum is exposure time times the gap-decline covariance, $(a - r + 1)(C_{Hd} + aV_d)$, where the level H in the gap contributes C_{Hd} and the accumulated slope $a d$ contributes aV_d . Starting later drops the low- a terms, where slope heterogeneity has had little time to open gaps, and so tilts the mix toward V_d .
- Treatment side: the parameters enter only through the total variance, so no decomposition is needed:

$$T(r) = 2 \sum_{a \geq r} \left[(1 - \bar{h}_a)^2 + \underbrace{V_H + 2aC_{Hd} + a^2V_d + v_a}_{= \text{Var}(h_a), \text{ directly observable}} \right]. \quad (32)$$

- Every case supplies all the inputs to (31) and (32): Cases 1 and 2 from three waves, Case 3 from four. In Cases 1-2 the inputs come from the diagonal, so repeated cross-sections suffice; Case 3 needs the first row and two interior cells, and so a panel.

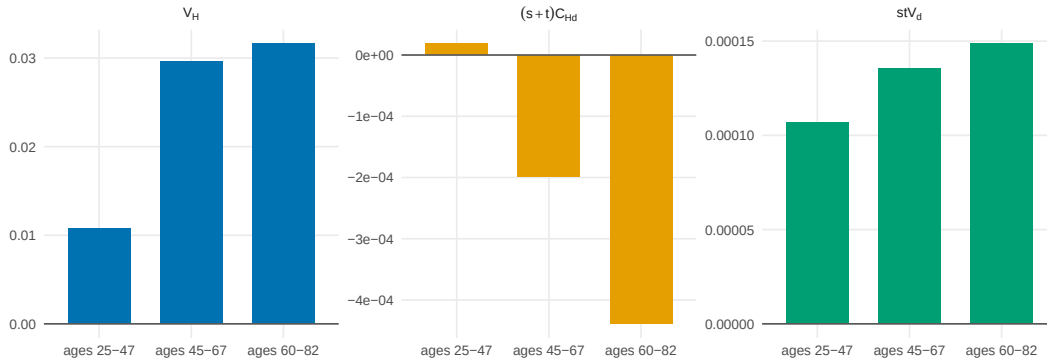
The local reading.

- Nothing above requires one line for life: A1 is linearity over the waves used, so the parameters are local to the window around the base age. Define the local level-change covariance $C_{Hd}(a) \equiv \text{Cov}_j(\bar{h}_{j,a}, d_j)$; within the window, $C_{Hd}(a) = C_{Hd} + a V_d$.

² $K(r)$ is not cheaper to identify than the pair: no two-wave combination is proportional to it, so it costs the same moments as (C_{Hd}, V_d) , and the timing profile needs the two separately. Appendix A states this and gives the weights $A(r), B(r)$.

- The curves of Section 3 read the local object directly. The one-step growth of the deterministic variance is $\text{Var}_j(\bar{h}_{j,a+1}) - \text{Var}_j(\bar{h}_{j,a}) = C_{Hd}(a) + C_{Hd}(a+1)$: the variance rises exactly while the local covariance is positive, and falls once it turns negative.
- A variance profile that turns over therefore requires $C_{Hd}(a)$ to change sign. The global linear model cannot produce it - there $C_{Hd}(a) = C_{Hd} + aV_d$ only increases. Figure 3 shows the cross-sectional version of this fact; how much of it is the local covariance itself and how much is composition - who survives to be measured at each age - is the empirical section's question, with a first pass parked in Appendix E.
- Beyond the reach of the noise, the row anchored at s falls or rises at rate $C_{Hd}(s)$; where the row runs past the window, its slope at lag k is the covariance of the base-age level with the change at the age traversed, $\text{Cov}_j(\bar{h}_{j,s}, d_{j,s+k})$ - the rows are forward-looking.
- The between-type channel of (31) is the exposure-weighted sum of the local objects: each summand is $(a - r + 1) C_{Hd}(a)$. The return never needed the global parameters as such; the linear window is one way of extrapolating the local sequence.
- Figure 11 is the local reading on the data: the denoising solve of the triple at three base-age windows. V_H grows as shocks accumulate into the level; the covariance term starts near zero and turns negative - the compression the rows of Figure 5 show directly.
- The sign of the small covariance term is where the noise treatment bites: the Case 2 read of Figure 8 holds the noise variance fixed and finds it positive in mid-life, while the denoising solve lets the noise move and finds it negative from mid-life on. Which is right depends on the noise process, and settling that is the empirical section's job.
- A first empirical pass at $C_{Hd}(a)$, and the estimation issues it raises, is parked in Appendix E.

Figure 11: The local triple in the data (primary measure): the denoising solve on twelve-year windows at three life stages. Each panel has its own scale.



- The returns sum to the end of the horizon, while the window identifies the local parameters. Carrying them forward is an extrapolation, and the linear window is its maximal-persistence case: the slope a person carries today is the slope they keep.
- If slopes mean-revert across windows, the compounding terms shrink, and the long-horizon return falls with the persistence of the slope. That persistence is an estimand, not a convention: it lives in the long-lag cells of the covariance matrix.

[With enough waves the cross-window persistence of d is estimable from long-lag interior cells; the returns section should eventually report long-horizon returns under the estimated persistence]

rather than the perfect-persistence default.]

Table 2 collects the dictionary from policy objects to the moments and data that deliver them.

Table 2: What each policy object needs

Policy object	Moments needed	Data
Healthcare cost; treatment return $T(r)$	mean path; $\text{Var}(h_a)$	cross-sections
Prevention return $P(r)$, fixed r	mean path; one combination of C_{Hd} and V_d	3 cross-sections if the noise variance is stationary; a 4-wave panel if it drifts
Timing profile $r \mapsto P(r)$	mean path; C_{Hd} and V_d separately	as above
Targeting on h_0 (Result 6)	first row; the level/noise split	panel
Targeting gain, type vs. uniform (App. B)	spread of P_j across types	panel + clusters
Targeting gain, type vs. level (App. B)	how well the pre-policy level predicts P_j ; noise in that level	panel + clusters

The two prevention rows need the same moments: identifying the combination is not cheaper than identifying the pair (Appendix A).

Obviously, a panel is still useful where cross-sections suffice - it tests the stationarity the cross-sectional reads rely on, and delivers the level/noise split that targeting needs.

[Given that the $K(r)$ combination needs as many moments to identify as C_{Hd} and V_d do, we could merge the two $P(r)$ rows of this table into one.]

The framework in the data. Return to Figure 3, with the section's machinery in hand.

- The variance rises steeply to about age 60 and then stops growing (the thin-celled tail past 80 is noisy), even as mean decline accelerates. Under a stationary noise variance this says the local covariance $C_{Hd}(a)$ has reached zero by 60, and the one-year spike moves too little to change that reading (Appendix E).
- The rows steepen with the base age, as in Figure 5: the old rows' sustained fall says the local covariance keeps falling past zero.
- One caution before reading these curves as pure dynamics: part of the variance flattening is composition - the tails are thinned by who survives to be measured - and the split between genuine compression and composition, with selection, floors and cohort effects handled properly, is the empirical companion's job. The framework's job here is only to show that its objects are visible, and readable, in the data.

A Derivations and proofs

Throughout, A2 makes (H_i, d_i) uncorrelated with all u terms, so $\text{Cov}(h_s, h_t) = V_H + (s+t)C_{Hd} + st V_d + \text{Cov}(u_s, u_t)$ as in (5).

Costs and the treatment return

Step 1: the cost of a random health level. For any random h , $\mathbb{E}[(1-h)^2] = (1 - \mathbb{E}h)^2 + \text{Var}(h)$.

Step 2: type and population levels. Conditioning on j and using $\mathbb{E}[h_{i,a} | j] = \bar{h}_{j,a}$ and $\text{Var}(h_{i,a} | j) = \text{Var}(u_{i,a} | j)$ gives the type-level cost; the law of total variance gives (3).

Step 3: the treatment return. Treatment scales each age- $a \geq r$ term by $(1-\tau)^2$; differentiating at $\tau = 0$ gives Result 1. $\square$

Prevention returns

Step 1: the type-level derivative. For $a \geq r$, $\mathbb{E}[\mathcal{C}_i | j]$ contains $(1 - \bar{h}_{j,a} - \mu(\bar{h}_{j,r-1} - \bar{h}_{j,a}))^2$ - prevention raises health, shrinking the gap - so $\partial/\partial\mu|_0 = -2(1 - \bar{h}_{j,a})(\bar{h}_{j,r-1} - \bar{h}_{j,a})$, giving Result 2.

Step 2: the population under a uniform policy. With the same μ for everyone, $\bar{h}_{j,a}(\mu) = \bar{h}_{j,a} + \mu(\bar{h}_{j,r-1} - \bar{h}_{j,a})$ and $\bar{h}_a(\mu) = \bar{h}_a + \mu(\bar{h}_{r-1} - \bar{h}_a)$. The mean-gap term contributes $\partial/\partial\mu|_0 (1 - \bar{h}_a(\mu))^2 = -2(1 - \bar{h}_a)(\bar{h}_{r-1} - \bar{h}_a)$.

Step 3: the between-type term. Using $\sum_j \pi_j (\bar{h}_{j,a} - \bar{h}_a) = 0$,

$$\frac{\partial}{\partial\mu}\bigg|_0 \sum_j \pi_j (\bar{h}_{j,a}(\mu) - \bar{h}_a(\mu))^2 = 2 \text{Cov}_j(\bar{h}_{j,a}, \bar{h}_{j,r-1} - \bar{h}_{j,a}). \quad (33)$$

Step 4: collecting.

$$\begin{aligned} P(r) &= 2 \sum_{a \geq r} \left[(1 - \bar{h}_a)(\bar{h}_{r-1} - \bar{h}_a) - \text{Cov}_j(\bar{h}_{j,a}, \bar{h}_{j,r-1} - \bar{h}_{j,a}) \right] \\ &= 2 \sum_{a \geq r} \left[(1 - \bar{h}_a)(\bar{h}_{r-1} - \bar{h}_a) + \text{Cov}_j(1 - \bar{h}_{j,a}, \bar{h}_{j,r-1} - \bar{h}_{j,a}) \right], \end{aligned} \quad (34)$$

and expanding the covariance gives the level-covariance form $\text{Var}_j(\bar{h}_{j,a}) - \text{Cov}_j(\bar{h}_{j,a}, \bar{h}_{j,r-1})$. For the decomposition (10), split $1 - \bar{h}_{j,a} = (1 - H_j) + (H_j - \bar{h}_{j,r-1}) + (\bar{h}_{j,r-1} - \bar{h}_{j,a})$ and apply covariance linearity. $\square$

Aggregation. $\mathbb{E}_j[(1 - \bar{h}_{j,a})(\bar{h}_{j,r-1} - \bar{h}_{j,a})] = (1 - \bar{h}_a)(\bar{h}_{r-1} - \bar{h}_a) + \text{Cov}_j(1 - \bar{h}_{j,a}, \bar{h}_{j,r-1} - \bar{h}_{j,a})$ confirms $P(r) = \sum_j \pi_j P_j(r)$.

Numerical check. Two types, $\pi_A = \pi_B = \frac{1}{2}$, one age; $\bar{h}_{A,a} = 0.9$ with no preventable decline, $\bar{h}_{B,a} = 0.3$ with preventable decline 0.3 (so $\bar{h}_a = 0.6$ and mean decline 0.15). Directly, $\mathbb{E}[\mathcal{C}_i(\mu)] = \frac{1}{2}[(0.1)^2 + (0.7 - 0.3\mu)^2]$ gives $P = 0.21$; formula (9) gives $2[0.06 + 0.045] = 0.21 \checkmark$. The pairing matters: using $\bar{h}_{j,a}$ in place of the gap $1 - \bar{h}_{j,a}$ would give 0.03.

The prevention combination $K(r)$

Result A.1 (What the prevention return needs). *In the linear case of Section 2.2:*

600 (i) For a fixed starting age r , $P(r)$ requires beyond the mean path only the single combination

$$K(r) \equiv \sum_{a=r}^{\bar{a}} (a-r+1)(C_{Hd} + aV_d) = A(r)C_{Hd} + B(r)V_d, \quad \begin{aligned} A(r) &\equiv \sum_{a=r}^{\bar{a}} (a-r+1), \\ B(r) &\equiv \sum_{a=r}^{\bar{a}} a(a-r+1), \end{aligned} \quad (35)$$

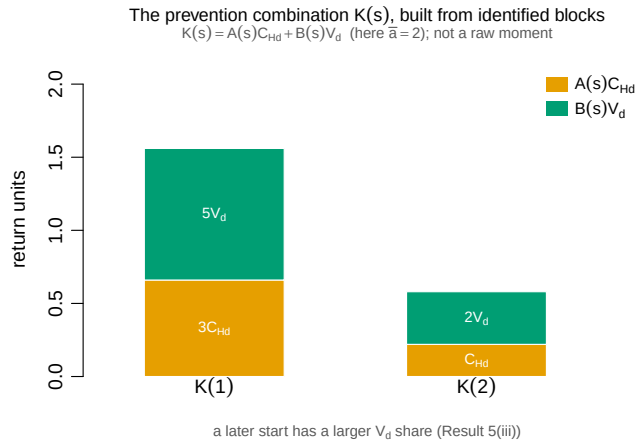
601 not C_{Hd} and V_d separately. The weights $A(r)$ and $B(r)$ are known once the remaining horizon
602 is fixed.

604 (ii) The profile $r \mapsto P(r)$ - the timing question - needs C_{Hd} and V_d separately, equivalently $K(r)$
605 at two starting ages. No new moments are needed beyond (i): the weight ratio $B(r)/A(r)$ -
606 the exposure-weighted average age of the remaining window - strictly increases in r , so no
607 two starting ages use the parameters in the same proportion and two K 's invert to the pair.

608 *Proof.* The linear case gives the between-type channel $(a-r+1)(C_{Hd} + aV_d)$ at each age
609 (Section 3.4); summing over $a \geq r$ yields $K(r)$, one linear functional of (C_{Hd}, V_d) whose weights
610 vary with r , so two values of r recover the pair. $\square$

- 611 • $K(r)$ is not cheaper to identify than the pair: no two-wave combination is proportional to
612 $K(r)$, so it costs the same moments as (C_{Hd}, V_d) . This is why the main text identifies the
613 pair and treats $K(r)$ as bookkeeping.
- 614 • A note on coordinates: V_d and every $K(r)$ are invariant to where age zero is placed, but
615 C_{Hd} is not - re-basing age by c shifts $C_{Hd} \mapsto C_{Hd} + cV_d$, since H_i is the level *at* the origin.
616 The profile needs two origin-free numbers; (C_{Hd}, V_d) is one convenient coordinate system for
617 them, given our origin at the first wave.
- 618 • Figure 12 shows $K(r)$ assembled from the identified blocks: the V_d share grows with the
619 starting age.

Figure 12: The prevention combination $K(r) = A(r)C_{Hd} + B(r)V_d$, for a horizon ending at $\bar{a} = 2$: $K(1) = 3C_{Hd} + 5V_d$ and $K(2) = C_{Hd} + 2V_d$. Block heights are illustrative.



620 When the diagonal is not enough

621 **Lemma A.1** (When the diagonal is not enough). (i) With the noise variances $\{v_a\}$ unrestricted,
622 the variance profile $\{\text{Var}(h_a)\}_a$ can never rule out pure noise, nor establish any deterministic

heterogeneity: pure noise ($V_H = C_{Hd} = V_d = 0$ with $v_a = \text{Var}(h_a)$) reproduces any profile exactly, and so does any (V_H, C_{Hd}, V_d) that is a valid second-moment matrix ($V_H, V_d \geq 0$, $C_{Hd}^2 \leq V_H V_d$) and whose implied deterministic variance $V_H + 2aC_{Hd} + a^2V_d$ lies weakly below the profile at every observed age. The diagonal bounds that deterministic variance from above at each age, never from below. (ii) Even with stationary noise variance, the diagonal never separates V_H from the noise: they enter every cell of it only as the bundle $V_H + \sigma_u^2$. Restoring identification requires stationarity restrictions on the noise for (i), and at least one off-diagonal cell for (ii).

Proof of Lemma A.1. (i) $\text{Var}(h_a) = V_H + 2aC_{Hd} + a^2V_d + v_a$. Given any profile $\{\text{Var}(h_a)\}$, set $V_H = C_{Hd} = V_d = 0$ and $v_a = \text{Var}(h_a) \geq 0$: the profile is reproduced exactly. More generally any (V_H, C_{Hd}, V_d) with $V_H + 2aC_{Hd} + a^2V_d \leq \text{Var}(h_a)$ for all observed a is consistent (valid meaning $V_H, V_d \geq 0$, $C_{Hd}^2 \leq V_H V_d$), with v_a the (nonnegative) remainder. (ii) With stationary σ_u^2 , profile differences identify (C_{Hd}, V_d) , but every diagonal cell is then $(V_H + \sigma_u^2) + \text{known terms}$: shifting $V_H \mapsto V_H + c$, $\sigma_u^2 \mapsto \sigma_u^2 - c$ leaves the whole diagonal unchanged. $\square$

Proposition 1, case by case

Case 1 (i.i.d. noise, stationary variance). Any noise with stationary variance gives $\text{Var}(h_a) = V_H + 2aC_{Hd} + a^2V_d + \sigma_u^2$, so first differences of the variance profile are $\text{Var}(h_a) - \text{Var}(h_{a-1}) = 2C_{Hd} + (2a - 1)V_d$ and the second difference is $2V_d$ - the reads (22). With $C_{Hd} = 0$ the first difference alone is V_d (the benchmark remark). For the level split, i.i.d. noise leaves off-diagonals clean: $\text{Cov}(h_0, h_1) = V_H + C_{Hd}$ gives V_H , and $\text{Var}(h_0) = V_H + \sigma_u^2$ gives σ_u^2 . Overidentification: every remaining cell equals its implied smooth value, e.g. $\text{Cov}(h_0, h_2) - \text{Cov}(h_0, h_1) = C_{Hd}$.

Case 2 (AR(1) noise, stationary variance). Stationarity gives $\text{Cov}(u_s, u_t) = \sigma_u^2 \rho^{t-s}$, hence the stated level covariances, and the diagonal reads carry over unchanged (only the stationary marginal variance enters the diagonal). With C_{Hd} known, the detilted row of the main text, $\tilde{C}_0(a) = \text{Cov}(h_0, h_a) - aC_{Hd} = V_H + \rho^a \sigma_u^2$ for $a = 0, 1, 2$ (reading $\tilde{C}_0(0) = \text{Var}(h_0)$), satisfies $\tilde{C}_0(0) - \tilde{C}_0(1) = (1 - \rho)\sigma_u^2$ and $\tilde{C}_0(1) - \tilde{C}_0(2) = \rho(1 - \rho)\sigma_u^2$: the ratio is ρ , and σ_u^2, V_H unwind. $\text{Cov}(h_1, h_2) = V_H + 3C_{Hd} + 2V_d + \rho\sigma_u^2$ is then implied - the overidentifying check.

Case 3 (AR(1) noise, age-specific variances). With $u_a = \rho u_{a-1} + \eta_a$ and free innovation variances, u_t equals $\rho^{t-s} u_s$ plus terms independent of u_s (for $t > s$), so $\text{Cov}(u_s, u_t) = \rho^{t-s} v_s$ with $v_s \equiv \text{Var}(u_s)$ free. Row 0 including its diagonal cell satisfies $\text{Cov}(h_0, h_k) = V_H + kC_{Hd} + \rho^k v_0$, so its second differences satisfy

$$S_k \equiv \text{Cov}(h_0, h_{k+1}) - 2\text{Cov}(h_0, h_k) + \text{Cov}(h_0, h_{k-1}) = v_0 \rho^{k-1} (1 - \rho)^2, \quad (36)$$

where S_k is local shorthand for the second differences written out in (27). This gives $\rho = S_2/S_1$ with four waves; then $v_0 = S_1/(1 - \rho)^2$, $V_H = \text{Var}(h_0) - v_0$, and $C_{Hd} = \text{Cov}(h_0, h_1) - \text{Var}(h_0) + (1 - \rho)v_0$. For V_d , take the two row-1 cells and eliminate v_1 :

$$V_d = \frac{\text{Cov}(h_1, h_3) - \rho \text{Cov}(h_1, h_2) - (1 - \rho)V_H - (4 - 3\rho)C_{Hd}}{3 - 2\rho}, \quad (37)$$

and the remaining $v_a = \text{Var}(h_a) - V_H - 2aC_{Hd} - a^2V_d$ come off the diagonals (their off-diagonal counterparts are then overidentifying checks). All verified in `check_framework.R`.

Case 4 (AR(1) noise plus one-period measurement error). With $u_a = w_a + \varepsilon_a$, w stationary AR(1) and ε i.i.d., $\text{Cov}(u_s, u_t) = \rho^{t-s} \sigma_w^2 + \mathbf{1}\{s = t\} \sigma_\varepsilon^2$: the spike adds to every diagonal cell and to no off-diagonal cell. Total noise variance is constant, so the diagonal reads of Case 1 deliver (C_{Hd}, V_d) unchanged. Detilting the row, $\tilde{C}_0(a) = V_H + \rho^a \sigma_w^2$ for $a \geq 1$ while $\tilde{C}_0(0) = \text{Var}(h_0) =$

$V_H + \sigma_w^2 + \sigma_\varepsilon^2$; successive drops among the off-diagonal cells are $\tilde{C}_0(a) - \tilde{C}_0(a+1) = \rho^a(1-\rho)\sigma_w^2$, so their ratio is ρ , the scale gives σ_w^2 , the level gives V_H , and $\sigma_\varepsilon^2 = \text{Var}(h_0) - V_H - \sigma_w^2$ - the reads (29). Three waves are not enough despite the count of six moments for six parameters: given the two diagonal differences, $\text{Cov}(h_1, h_2) - \text{Cov}(h_0, h_1) = 2C_{Hd} + 2V_d$ is implied, so only three independent equations remain for the four unknowns $(V_H, \rho, \sigma_w^2, \sigma_\varepsilon^2)$ - the map is rank-deficient, and the fourth wave is needed. Interior cells are then overidentifying checks, e.g. $\text{Cov}(h_1, h_2) = V_H + 3C_{Hd} + 2V_d + \rho\sigma_w^2$. Applied blind to this world, Case 2's fade read uses the contaminated first drop $\tilde{C}_0(0) - \tilde{C}_0(1) = (1-\rho)\sigma_w^2 + \sigma_\varepsilon^2$ and understates ρ badly. The five-wave general member (free v_s and free $\sigma_{\varepsilon,s}^2$) works the same way one row down: second differences of the row *excluding the anchor* give ρ and v_0 , first differences give C_{Hd} , then V_H , the spike at each age as the diagonal remainder, and V_d from the interior as in Case 3. All verified in `check_framework.R`, including exact recovery on population moments.

The denoising case (Appendix D). With independent noise and free v_a , off-diagonal cells are pure smooth layer; along row a the smooth layer is linear in the other index, so its diagonal value is the average of its two neighbours,

$$\frac{1}{2}[\text{smooth}(a-1, a) + \text{smooth}(a, a+1)] = V_H + 2aC_{Hd} + a^2V_d = \text{smooth}(a, a), \quad (38)$$

which is the interior formula for $\text{Var}^*(h_a)$; at the edges extrapolate along row 0 or column 2, e.g. $\text{Var}^*(h_0) = 2\text{Cov}(h_0, h_1) - \text{Cov}(h_0, h_2)$ and $\text{Var}^*(h_2) = 2\text{Cov}(h_1, h_2) - \text{Cov}(h_0, h_2)$. The starred profile is the noise-free variance profile, so the Case 1 reads applied to it recover (V_H, C_{Hd}, V_d) , and $v_a = \text{Var}(h_a) - \text{Var}^*(h_a)$. Expanding the starred formulas reproduces the covariance-only solve ($C_{Hd} = \text{Cov}(h_0, h_2) - \text{Cov}(h_0, h_1)$, etc.) term by term.

The D_k ladder

The ladder is a diagnostic rather than a step in any proof: it reads the noise process off changes instead of levels. With $\gamma(k) = \sigma_u^2 \rho^k$,

$$\text{Cov}(\Delta u_t, \Delta u_{t+k}) = 2\gamma(k) - \gamma(k-1) - \gamma(k+1) = -\sigma_u^2(1-\rho)^2 \rho^{k-1}, \quad k \geq 1, \quad (39)$$

so $D_k = V_d - (1-\rho)^2 \rho^{k-1} \sigma_u^2$: dip-then-flat at $\rho = 0$, a geometric rise for $\rho \in (0, 1)$, flat at $\rho = 1$. Note that dip-then-flat is shared by the random-walk-plus-spike class of Proposition D.1 ($D_1 = V_d - \sigma_\varepsilon^2$, $D_k = V_d$ for $k \geq 2$), so the ladder alone does not separate i.i.d. noise from that class - the level staircase does. Successive gaps give $\rho = (D_3 - D_2)/(D_2 - D_1)$ (five waves). An added i.i.d. spike ε changes no D_k with $k \geq 2$ (adjacent changes alone share an ε) but shifts D_1 by $-\sigma_\varepsilon^2$, so the five-wave formula is not spike-robust either: the spike-robust recovery uses gaps among the $k \geq 2$ rungs only, $\rho = (D_4 - D_3)/(D_3 - D_2)$, at the cost of a sixth wave. The three-wave fade solve fails under a spike for the same reason (the spike enters the detilted row at $a = 0$ but none of its off-diagonal cells); Case 4 is the level-based repair, moving the fade read to drops the spike cannot touch at the cost of a fourth wave. The simulation check in `check_framework.R` ($N = 2 \times 10^6$) recovers all parameters to two-three decimals under both spike-free routes.

B The value of targeting: pricing information sets

This appendix proves the targeting results of Section 2.6 and adds the comparison between type- and level-targeting. Throughout, $\mu(z)$ is an allocation rule based on information z , with average intensity held fixed at $\mathbb{E}[\mu(z)] = \bar{\mu}$.

- Result 5 is the accounting identity: any rule saves exactly $\text{Cov}(\mu(z), \mathbb{E}[P_i | z])$ over the uniform policy. The optimality counterpart is Lemma B.1: per unit of intensity dispersion, no rule based on z beats $\text{sd}(\mathbb{E}[P_i | z])$.
- Result 7's third-moment point, formally: $P_j(r) = 2 \sum_{a \geq r} (1 - H_j - a d_j) (-(a - r + 1) d_j)$ is quadratic in (H_j, d_j) , so the spread of P_j across types - the object any type-based scheme collects - involves third and fourth moments of the type distribution, beyond the covariance surface of Section 3. This is the sense in which targeting is priced by the type distribution (clusters), not by second moments.

Lemma B.1 (Value of an information set). *To first order in intensities, the saving of allocation $\mu(z)$ over the uniform policy with the same average intensity is $\text{Cov}(\mu(z), \mathbb{E}[P_i | z])$ (Result 5). Per unit of intensity dispersion $\text{sd}(\mu)$, no rule based on z can save more than $\text{sd}(\mathbb{E}[P_i | z])$, with equality when μ is increasing and affine in $\mathbb{E}[P_i | z]$. An information set is worth the return dispersion it reveals. (proof below.)*

Result B.1 (Targeted vs. uniform prevention; $z = \text{type}$). *Bound for Result 7: the saving from any type-based allocation is $\text{Cov}_j(\mu_j, P_j(r)) \leq \text{sd}(\mu) \text{sd}_j(P_j(r))$; it is zero iff $P_j(r)$ is constant across types. The opportunity cost of not targeting is the foregone covariance.*

Result B.2 (Type- vs. level-targeting; $z = h_{i,r-1}$). *Let the level signal be the last pre-intervention observation $h_{i,r-1}$. If u is i.i.d., future gaps depend on z only through the type, so $\mathbb{E}[P_i | h_{i,r-1}] = \mathbb{E}[P_j | h_{i,r-1}]$ and*

$$\frac{\text{gain from level-targeting}}{\text{gain from type-targeting}} = \frac{\text{sd}(\mathbb{E}[P_j | h_{i,r-1}])}{\text{sd}_j(P_j)} \leq 1, \quad (\text{linear rules: } |\text{Corr}(h_{i,r-1}, P_j)|). \quad (40)$$

- Two forces keep the ratio below one: (i) *span* - a single level reveals only one combination, $H + (r - 1)d$ plus noise, of the two traits (H, d) that drive P_j ; (ii) *attenuation* - v_{r-1} in $\text{Var}(h_{r-1})$ shrinks the projection. The value of clustering is the residual dispersion, largest when current level poorly predicts subsequent decline - and directly computable once clusters are estimated (compute P_j per cluster, project on the pre-period level).
- In the linear case the two forces factor cleanly: writing $h_{i,r-1} = \bar{h}_{i,r-1} + u_{i,r-1}$, where $\bar{h}_{i,r-1} = H_i + (r - 1)d_i$ is the individual's expected (deterministic) health,

$$\begin{aligned} \text{Corr}(h_{r-1}, P_j) &= \sqrt{\lambda_{r-1}} \text{Corr}(\bar{h}_{i,r-1}, P_j), \\ \lambda_{r-1} &\equiv \frac{\text{Var}(\bar{h}_{i,r-1})}{\text{Var}(h_{r-1})} = \frac{V_H + 2(r - 1)C_{Hd} + (r - 1)^2 V_d}{V_H + 2(r - 1)C_{Hd} + (r - 1)^2 V_d + v_{r-1}}, \end{aligned} \quad (41)$$

a signal-reliability factor $\sqrt{\lambda_{r-1}}$ (attenuation, computable from Proposition 1) times a span factor $\text{Corr}(\bar{h}_{i,r-1}, P_j)$ (generically < 1 : one linear combination of two traits).

- If u is persistent (random walk), level-targeting gains a genuine extra role: deviations persist into future levels and raise individual returns through the gap term. If u is mean-reverting, levels *mislead*: dips predict recovery. Both cases reinforce that the deterministic component is the reliable targeting object.

Notation for the proofs: $\mathcal{C}_i$ is individual i 's lifetime cost (Section 2), P_i their marginal return to prevention (the reduction in expected lifetime cost per unit of intensity), and z_i the information the allocation may condition on; the predicted return is written $\mathbb{E}[P_i | z]$ throughout.

Proof of Lemma B.1

Step 1: linearize in intensities. Giving individual i intensity μ_i lowers their expected cost by $\mu_i P_i$ to first order (that is the definition of the marginal return) so for any allocation rule $\mu(\cdot)$,

$$\mathbb{E}[\mathcal{C}_i(\mu(\cdot))] = \mathbb{E}[\mathcal{C}_i(0)] - \mathbb{E}[\mu(z_i) P_i] + O(\mu^2). \quad (42)$$

Step 2: only the predicted return matters. The rule can respond to z_i only, so by the law of iterated expectations,

$$\mathbb{E}[\mu(z_i) P_i] = \mathbb{E}[\mu(z_i) \mathbb{E}(P_i | z_i)]. \quad (43)$$

Step 3: split the effect into budget and targeting.

$$\mathbb{E}[\mu(z) \mathbb{E}[P_i | z]] = \underbrace{\bar{\mu} \mathbb{E}[P_i]}_{\text{uniform policy, same budget}} + \underbrace{\text{Cov}(\mu(z), \mathbb{E}[P_i | z])}_{\text{gain from targeting}}, \quad (44)$$

so the cost reduction of $\mu(z)$ over uniform $\bar{\mu}$ is exactly the covariance, as stated in the lemma.

Step 4: the bound. By Cauchy-Schwarz, $\text{Cov}(\mu, \mathbb{E}[P_i | z]) \leq \text{sd}(\mu) \text{sd}(\mathbb{E}[P_i | z])$, with equality iff μ is an increasing affine transform of $\mathbb{E}[P_i | z]$:

$$\mu^*(z) = \bar{\mu} + \text{sd}(\mu) \frac{\mathbb{E}[P_i | z] - \mathbb{E}[P_i]}{\text{sd}(\mathbb{E}[P_i | z])}, \quad \text{saving} = \text{sd}(\mu) \text{sd}(\mathbb{E}[P_i | z]). \quad (45)$$

Feasibility of the linear rule

The rule $\mu^*(z)$ stays in $[0, 1]$ for every z iff

$$\text{sd}(\mu) \leq \min(\bar{\mu}, 1 - \bar{\mu}) \frac{\text{sd}(\mathbb{E}[P_i | z])}{\max_z |\mathbb{E}[P_i | z] - \mathbb{E}[P_i]|}, \quad (46)$$

which holds automatically for small $\text{sd}(\mu)$ whenever returns are bounded - in particular for finitely many types, so Result B.1 needs no caveat. When $\mathbb{E}[P_i | z]$ is unbounded (e.g. a Gaussian level signal), clip the rule to $[0, 1]$. Writing $S = (\mathbb{E}[P_i | z] - \mathbb{E}[P_i])/\text{sd}(\mathbb{E}[P_i | z])$, clipping binds only on the tail event $|S| > \min(\bar{\mu}, 1 - \bar{\mu})/\text{sd}(\mu)$ and forgoes saving $\text{sd}(\mu) \text{sd}(\mathbb{E}[P_i | z]) \mathbb{E}[S^2; |S| > \min(\bar{\mu}, 1 - \bar{\mu})/\text{sd}(\mu)]$ - exponentially small in $1/\text{sd}(\mu)$ for Gaussian signals - so the stated saving is exact to leading order.

Alternative: box constraints and bang-bang targeting

Impose $\mu(z) \in [0, 1]$ instead of fixing the dispersion of μ , keeping the budget $\mathbb{E}[\mu] = \bar{\mu}$. The objective is linear in $\mu(z)$ pointwise, so the optimum is a corner solution: $\mu = 1$ where $\mathbb{E}[P_i | z]$ exceeds a threshold and $\mu = 0$ below it, with the threshold set so the treated group has mass $\bar{\mu}$ - “treat the top $\bar{\mu}$ quantile”. The gain over uniform is the upper-tail object

$$\bar{\mu} (\mathbb{E}[P_i | \text{top } \bar{\mu}] - \mathbb{E}[P_i]). \quad (47)$$

If $\mathbb{E}[P_i | z]$ is Gaussian, the truncated-normal mean $\mathbb{E}[P_i | \text{top } q] = \mathbb{E}[P_i] + \text{sd}(\mathbb{E}[P_i | z]) \varphi(\Phi^{-1}(1 - q))/q$ turns this into $\varphi(\Phi^{-1}(1 - \bar{\mu})) \text{sd}(\mathbb{E}[P_i | z])$: proportional to the same dispersion, with a factor that depends only on the budget - hence identical comparisons across information sets as under the affine rule.

Remark (budget devices). Fixing the dispersion of μ leads to an affine rule; the box constraint leads to treating the top- $\bar{\mu}$ quantile. Either way the saving is a budget factor, common to every information set, times $\text{sd}(\mathbb{E}[P_i | z])$ (for the box case, exactly so under Gaussian signals). Nothing in the text depends on which device is used: every comparison in Results B.1-B.2 and Result 6 is a ratio or a difference in which the budget factor cancels.

Proof of Result B.2

The level signal is the last pre-intervention observation $h_{i,r-1}$, and u is i.i.d.

Step 1: the level signal predicts returns only through the type. For $a \geq r$, $\text{Cov}(u_{r-1}, u_a) = 0$, so conditional on (j, h_{r-1}) the expected future gaps equal the type gaps $1 - \bar{h}_{j,a}$. Hence $P_i = P_j$, and $\mathbb{E}[P_i | h_{r-1}] = \mathbb{E}[P_j | h_{r-1}]$.

Step 2: apply the lemma to each information set. Per unit of intensity dispersion, type-targeting saves $\text{sd}_j(P_j)$; level-targeting saves $\text{sd}(\mathbb{E}[P_j | h_{r-1}])$.

Step 3: the ratio is below one. Conditioning can only shrink variance (law of total variance): $\text{Var}(\mathbb{E}[P_j | h_{r-1}]) \leq \text{Var}_j(P_j)$, giving the stated ratio.

Step 4: linear rules give the correlation form. For $\mu = \bar{\mu} + b h_{r-1}$,

$$\frac{\text{saving}}{\text{intensity dispersion}} = \frac{b \text{Cov}(h_{r-1}, P_j)}{|b| \text{sd}(h_{r-1})} = \pm \text{Corr}(h_{r-1}, P_j) \text{sd}(P_j), \quad (48)$$

so the best linear rule saves $|\text{Corr}(h_{r-1}, P_j)| \text{sd}(P_j)$ per unit of dispersion.

Step 5: reliability factorization (as used in the discussion of Result B.2). Since u_{r-1} is uncorrelated with (H_i, d_i) and P_j is a function of them, $\text{Cov}(h_{r-1}, P_j) = \text{Cov}(\bar{h}_{i,r-1}, P_j)$ while $\text{Var}(h_{r-1}) = \text{Var}(\bar{h}_{i,r-1}) + v_{r-1}$. Hence $\text{Corr}(h_{r-1}, P_j) = \sqrt{\lambda_{r-1}} \text{Corr}(\bar{h}_{i,r-1}, P_j)$ with $\lambda_{r-1} = \text{Var}(\bar{h}_{i,r-1})/\text{Var}(h_{r-1})$.

Proof of Results 6 and 7

Write $C_0 \equiv \text{Cov}(h_{i,0}, P_i(r))$ for the covariance in Result 6.

Step 1: noise in the signal drops out. $P_i = P_{j(i)}$ is a function of the type alone (Result 2) and $\mathbb{E}[u_{i,0} | j] = 0$, so $\text{Cov}(u_{i,0}, P_i) = \mathbb{E}[P_j \mathbb{E}[u_{i,0} | j]] = 0$ and $C_0 = \text{Cov}_j(\bar{h}_{j,0}, P_j)$. Likewise $\text{Var}(h_{i,0}) = \text{Var}_j(\bar{h}_{j,0}) + v_0$ with $v_0 \equiv \mathbb{E}_j[\text{Var}(u_{i,0} | j)]$.

Step 2: the projection slope. By definition $\beta(r) = \text{Cov}(h_{i,0}, P_i)/\text{Var}(h_{i,0})$, which with Step 1 is (15).

Step 3: the two channels. At each age P_j pairs the gap $Y_a \equiv 1 - \bar{h}_{j,a}$ with the preventable decline $Z_a \equiv \bar{h}_{j,r-1} - \bar{h}_{j,a}$. For any three variables, $\text{Cov}(X, YZ) = \mathbb{E}[Y] \text{Cov}(X, Z) + \mathbb{E}[Z] \text{Cov}(X, Y) + \mathbb{E}[\tilde{X} \tilde{Y} \tilde{Z}]$ with tildes denoting deviations from means. Apply this with $X = \bar{h}_{j,0}$: the means are $\mathbb{E}_j[Y_a] = 1 - \bar{h}_a$ and $\mathbb{E}_j[Z_a] = \bar{h}_{r-1} - \bar{h}_a$, and $\text{Cov}_j(\bar{h}_{j,0}, Y_a) = -\text{Cov}_j(\bar{h}_{j,0}, \bar{h}_{j,a})$. Summing over $a \geq r$ and dropping the third central moments gives (16).

Step 4: the linear case. With $\bar{h}_{j,a} = H_j + a d_j$ we have $\text{Cov}_j(\bar{h}_{j,0}, \bar{h}_{j,a}) = V_H + a C_{Hd}$ and $Z_a = -(a - r + 1) d_j$, so $\text{Cov}_j(\bar{h}_{j,0}, Z_a) = -(a - r + 1) C_{Hd}$; substituting into (16) and collecting the exposure weight $(a - r + 1)$ gives (17). Carrying out the sums with $A(r)$, $B(r)$ of (35) gives the equivalent closed form $2A(r) \bar{d} V_H + 2C_{Hd} [A(r)(\bar{H} - 1) + 2B(r) \bar{d}]$.

Step 5: per-person comparisons. A person one standard deviation below the mean has predicted return $\mathbb{E}[P_i] - \beta(r) \text{sd}(h_0)$, which under $\beta(r) < 0$ lies $|\beta(r)| \text{sd}(h_0)$ above the mean. Moving one unit of intensity from a person one sd above to one one sd below changes cost

by the difference in their predicted returns, $2|\beta(r)|\text{sd}(h_0)$ (Result 5), leaving average intensity unchanged.

Step 6: attenuation. If the type's own level $\bar{h}_{j,0}$ were observed, the projection of P_i on it would have slope $C_0/\text{Var}_j(\bar{h}_{j,0})$ - same covariance, smaller variance - so $\beta(r)$ is that slope times $\lambda_0 = \text{Var}_j(\bar{h}_{j,0})/(\text{Var}_j(\bar{h}_{j,0}) + v_0)$. The per-sd comparison scales with $|C_0|/\text{sd}(\cdot)$ and $\text{sd}_j(\bar{h}_{j,0})/\text{sd}(h_0) = \sqrt{\lambda_0}$, giving the $\sqrt{\lambda_0}$ factor.

Step 7: the ceiling (Result 7). The per-person statements are Result 5 with z the type. For the ceiling, $\mathbb{E}[P_i | z] = \mathbb{E}[\mathbb{E}[P_i | j] | z] = \mathbb{E}[P_j | z]$, so by the law of total variance $\text{Var}(\mathbb{E}[P_j | z]) \leq \text{Var}_j(P_j)$, with equality iff z determines P_j . $\square$

C Growth-regression representation: equivalence and corrections

Every case of Proposition 1 can equivalently be expressed in terms of individual growth regressions: OLS of $h_{i,a}$ on $(1, a)$, keeping intercepts $\hat{H}_i$, slopes $\hat{d}_i$ and residuals e_i . Three things are worth separating.

- The representation is general. Writing $(\hat{H}_i, \hat{d}_i, e_i) = Mh_i$, the matrix M depends only on which waves are used, not on the noise process, and is invertible; the coefficient second moments are therefore the congruence $M \text{Var}(h) M'$ of the level covariance matrix, whatever the noise. The two coordinate systems always carry identical information.
- The dictionary is case-specific. With $\hat{\theta}_i = \theta_i + Au_i$ and $e_i = Ru_i$, the noise enters as $A\Sigma_u A'$, $A\Sigma_u R'$ and $R\Sigma_u R'$. Under i.i.d. noise these collapse to $\sigma_u^2(X'X)^{-1}$ and $\sigma_u^2 RR'$, giving the dictionary below. Under memory they involve ρ as well, and under age-specific variances the design weights the individual v_a : the corrections change, the transform does not.
- Numerical equivalence needs just-identification. When both routes solve the same just-identified system they return identical estimates in any sample. With overidentifying moments and a fixed weight matrix, the two routes agree only if the weighting is mapped through the same transform.

Two practical caveats. The design must match the deterministic basis, so if a quadratic age term is added (Section 2.2) the regression becomes OLS on $(1, a, a^2)$ and the same argument goes through with three coefficients. Residual coordinates exist only from three waves on; with two waves the regression fits exactly and the noise correction must come from an off-diagonal cell. This is also why the empirical-Bayes shortcut at the end of this appendix fails under memory: the failure is in the noise correction, not in the representation. The congruence is verified under all three noise processes in `check_framework.R`.

Equivalence

Fix waves $a = 0, \dots, T-1$ and the design $X = [\mathbf{1}, a]$. Individual OLS gives $(\hat{H}_i, \hat{d}_i)' = (X'X)^{-1}X'h_i$; let e_i collect the residual coordinates (any basis of the orthogonal complement of $\text{col}(X)$, dimension $T-2$). Then $(\hat{H}_i, \hat{d}_i, e_i) = Mh_i$ with M invertible, so the second-moment matrix of $(\hat{H}, \hat{d}, e)$ is the congruence $M \text{Var}(h) M'$ of the level covariance matrix: the two coordinate systems carry *identical* information. Moreover the model's moment equations are *linear* in (V_H, V_d, C_{Hd}, v_a) and, with three waves, just-identified with a unique solution; hence any valid solving route - level moments (the denoising solve of Appendix D) or coefficient moments (below)

- returns *numerically identical* estimates in any sample. The two-wave benchmark ($C_{Hd} = 0$) is the special case where the regression fits exactly, $\hat{H}_i = h_{i,0}$ and $\hat{d}_i = \Delta h_{i,1}$, and the noise correction comes from $\text{Cov}(h_0, h_1)$.

Three-wave dictionary (Case 1: i.i.d. noise)

With $a = 0, 1, 2$:

$$\hat{d}_i = \frac{h_{i,2} - h_{i,0}}{2} = d_i + \frac{u_{i,2} - u_{i,0}}{2}, \quad \hat{H}_i = \frac{5h_{i,0} + 2h_{i,1} - h_{i,2}}{6} = H_i + \frac{5u_{i,0} + 2u_{i,1} - u_{i,2}}{6}, \quad (49)$$

$$e_i = h_{i,0} - 2h_{i,1} + h_{i,2} = u_{i,0} - 2u_{i,1} + u_{i,2} \quad (\text{the deterministic part is annihilated exactly}). \quad (50)$$

Second moments:

$$\text{Var}(\hat{d}) = V_d + \frac{v_0 + v_2}{4}, \quad \text{Cov}(\hat{H}, \hat{d}) = C_{Hd} - \frac{5v_0 + v_2}{12}, \quad \text{Var}(\hat{H}) = V_H + \frac{25v_0 + 4v_1 + v_2}{36}, \quad (51)$$

$$\text{Var}(e) = v_0 + 4v_1 + v_2, \quad \text{Cov}(\hat{d}, e) = \frac{v_2 - v_0}{2}, \quad \text{Cov}(\hat{H}, e) = \frac{5v_0 - 4v_1 - v_2}{6}. \quad (52)$$

The residual block (second row) identifies (v_0, v_1, v_2) ; substituting into the first row corrects $\text{Var}(\hat{d})$ and $\text{Cov}(\hat{H}, \hat{d})$ and reproduces the denoising solve of Appendix D exactly.

Interpretation. The first row is where the regression version earns its keep: V_d is *observed slope dispersion minus estimation noise* (the empirical-Bayes/shrinkage correction of the value-added and firm-effects literatures), and C_{Hd} is the intercept-slope covariance purged of the *mechanical negative correlation* from shared noise (a low baseline draw lowers $\hat{H}_i$ and raises $\hat{d}_i$).

The EB shortcut and its biases

EB = empirical Bayes: since estimated slopes are noisy, their raw dispersion overstates V_d by the average squared standard error, so the textbook practice (as in the teacher value-added and firm-effects literatures) subtracts it - the variance-estimation step of an empirical-Bayes shrinkage procedure. Concretely, the shortcut estimates $\hat{\sigma}_i^2 = e_i^2/6$ (RSS over $T - 2 = 1$ d.f.), $\widehat{SE}(\hat{d}_i)^2 = \hat{\sigma}_i^2/2$, and corrects $\hat{V}_d^{EB} = \text{Var}(\hat{d}_i) - \mathbb{E}[\hat{\sigma}_i^2]/2$. This replaces the three residual-block equations with one pooled homoskedastic equation, and is *not* the same statistic:

- i.i.d. noise, age-varying variance: bias = $\frac{v_0 - 2v_1 + v_2}{6}$ (zero iff v_a is linear in age; can push the estimate negative);
- random walk + i.i.d. noise (constant variances): bias = $+\sigma_\eta^2/3$ - permanent innovations masquerade as slope heterogeneity (the HIP/RIP confound).

Simulation check

$N = 2 \times 10^6$, four waves, true $V_d = 0.040$, $C_{Hd} = 0.050$ (`check_framework.R`):

Noise scenario	$\hat{V}_d$ (true 0.040)				$\hat{C}_{Hd}$ (true 0.050)
	denoise	growth-reg.	EB shortcut	Prop. D.1	denoise = g-reg.
i.i.d., homoskedastic	0.0401	0.0401	0.0402	0.0399	0.0503
i.i.d., age-varying v_a	0.0421	0.0421	-0.0250	0.0389	0.0484
random walk + noise	0.1406	0.1406	0.1065	0.0389	0.0487

886 The denoising solve and its growth-regression twin agree to $\leq 4 \times 10^{-14}$ in every scenario -
 887 including the third, where both are biased *identically* (the same statistic in different coordinates).
 888 The EB biases match the formulas above ($0.040 - 0.067 = -0.027$; $0.040 + 0.067 = 0.107$), and
 889 the denoising solve's random-walk bias is $+\sigma_\eta^2/2 = 0.100$, from $\text{Cov}(h_1, h_2)$ absorbing $\sigma_{\eta,1}^2$. With
 890 $T > 3$ waves both systems are overidentified in the same way (the bijection preserves the moment
 891 count); the lag-moment combinations of Case 2 and Proposition D.1 remain natural ones for
 892 serial dependence.

893 D The general covariance surface

894 The cases of Proposition 1 are cells of one template, stated fully generally in (54). One noise
 895 class deserves its own display - “random walk (η , initial variance ω^2) + i.i.d. (ε)”, the class of
 896 Proposition D.1 below (the $\rho \rightarrow 1$ member of the Case 3 family (25) with a spike added) - for
 897 which every second moment of the panel is one equation:

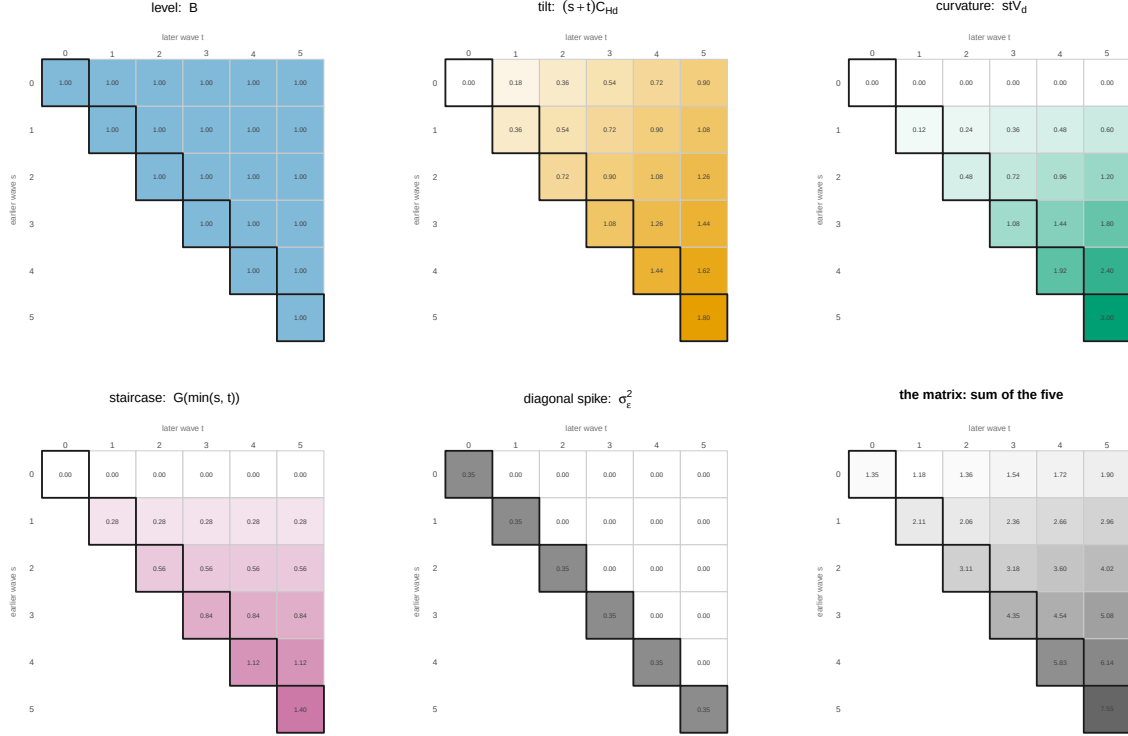
$$898 \quad \text{Cov}(h_s, h_t) = \underbrace{B}_{\text{level}} + \underbrace{(s+t)C_{Hd}}_{\text{tilt}} + \underbrace{stV_d}_{\text{curvature}} + \underbrace{G(\min(s,t))}_{\text{staircase}} + \underbrace{\mathbf{1}\{s=t\}\sigma_{\varepsilon,s}^2}_{\text{diagonal spike}}, \quad (53)$$

899 with $B \equiv V_H + \omega^2$ and $G(m) \equiv \sum_{\ell \leq m} \sigma_{\eta,\ell}^2$, $G(0) = 0$; and fully generally, for any deterministic
 900 basis $f_1, \dots, f_K$ and any noise process,

$$901 \quad \text{Cov}(h_s, h_t) = \sum_{k,l} \Omega_{kl} f_k(s) f_l(t) + \gamma_u(s, t), \quad (54)$$

902 where Ω is the second-moment matrix of the deterministic traits and γ_u the noise autocovariance.
 903 Our linear case is $f = (1, a)$; Case 2 is $\gamma_u(s, t) = \sigma_u^2 \rho^{t-s}$; Case 3 is $\gamma_u(s, t) = \rho^{t-s} v_s$ (Figure 14
 904 shows the menu of layers).

Figure 13: The covariance matrix, layer by layer (colours as in the anatomy charts)



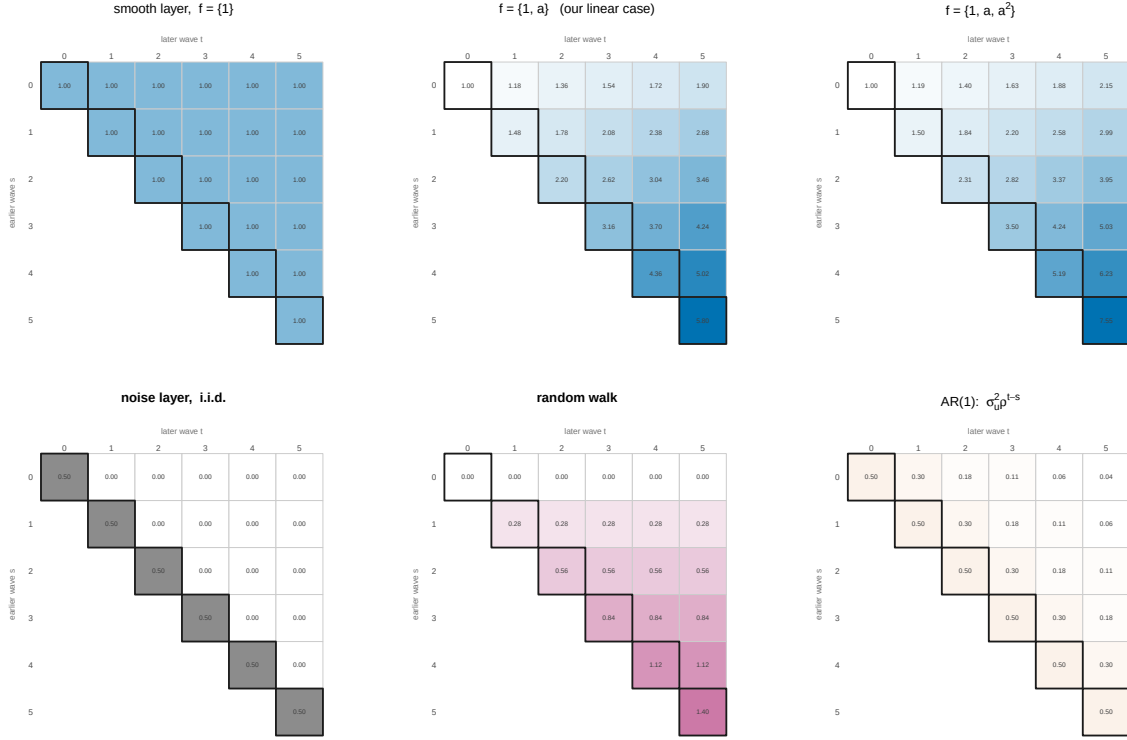
- Where the terms come from (Figure 13): the deterministic part contributes the polynomial layers, because $(H_i + s d_i)(H_i + t d_i) = H_i^2 + (s + t)H_i d_i + s t d_i^2$ - the product of two trend lines, multiplied out. The noise contributes the local layers: the walk shared up to the earlier wave ($\min(s, t)$), and the i.i.d. part only where a wave meets itself.
- The triad of Section 1.2 maps onto the layers: *stays* = level + staircase, *the moment* = the spike, *evolving* = tilt + curvature. The forces that keep these apart are in Section 3.2; the boundary case is a “noise” whose autocovariance were proportional to st , which would be an individual drift, i.e. it *is* a deterministic slope.

913 **Vocabulary**

	Term here	Meaning on the matrix	Standard counterpart
	surface	$\text{Cov}(h_s, h_t)$ as height over (s, t)	“covariance structure” (earnings dynamics); “covariance function/surface” (functional data analysis)
	level B	constant height everywhere	fixed-effect variance (σ_α^2), bundled with initial permanent variance
914	tilt $(s + t)C_{Hd}$	plane sloping up toward late-late pairs	intercept-slope covariance $\sigma_{\alpha\beta}$ in HIP earnings models
	curvature stV_d	twist of the surface: the cross-derivative $\partial^2/\partial s \partial t$	slope variance σ_β^2 ; “heterogeneous profiles”
	staircase $G(\min(s, t))$	terraces rising toward the diagonal	the unit-root signature: covariances growing in $\min(s, t)$ (the unit-root literature; see the companion note)
	diagonal spike σ_ϵ^2	a ridge only where $r = t$	measurement error; the “nugget effect” in geostatistics

- 915 • The spike is not an extra ingredient: it is the i.i.d. component already in the main text -
916 the entire noise in Case 1. Whether it is measurement error or genuine transitory health
917 variation cannot be distinguished from second moments (both paint only the diagonal), but
918 the interpretation matters for welfare: measurement error should be *excluded* from $\text{Var}(h_a)$
919 in the treatment return $T(r)$, while the prevention return $P(r)$ never contains the spike and
920 so is immune to the question - the caveat in Section 2.5.
- 921 • The earnings-dynamics ancestry of every object here, with full references, is documented in
922 the companion note `earnings_comparison_note.tex`.

Figure 14: The general template as a menu: pick one smooth layer, add noise layers



Age-varying noise without memory: the denoising case

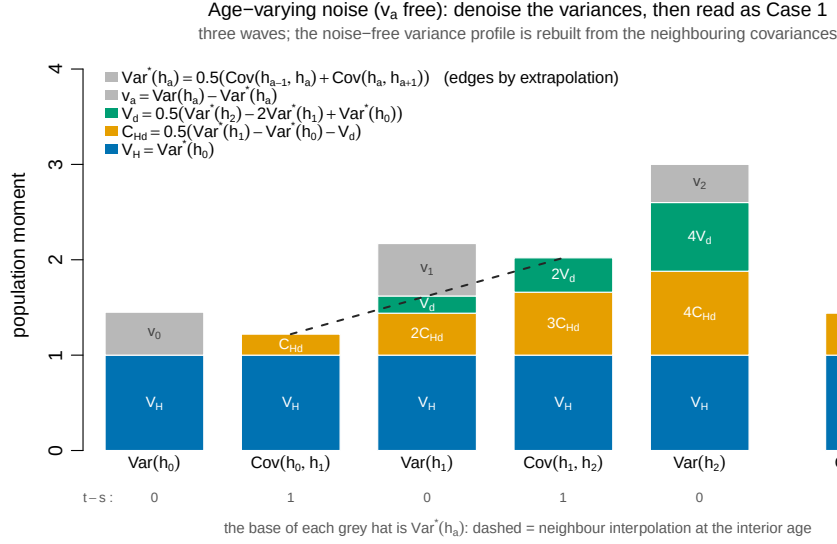
Case 3 handles age-specific variances with memory. The memoryless version ($\rho = 0$ with free v_a) is simpler and has its own device.

Independent noise, free v_a (the denoising case). Now no formula for (V_H, C_{Hd}, V_d) can put weight on any raw $\text{Var}(h_a)$: each variance carries its own free v_a that no other moment contains. But off-diagonal cells are pure smooth layer, and because the smooth layer is linear along each row, the *noise-free variance profile* can be rebuilt from them:

$$\text{Var}^*(h_a) = \frac{1}{2} [\text{Cov}(h_{a-1}, h_a) + \text{Cov}(h_a, h_{a+1})] \quad (\text{interior } a; \text{ edges by extrapolation}), \quad (55)$$

after which everything is Case 1 run on the starred profile (Figure 15): $v_a = \text{Var}(h_a) - \text{Var}^*(h_a)$, $V_d = \frac{1}{2} [\text{Var}^*(h_2) - 2\text{Var}^*(h_1) + \text{Var}^*(h_0)]$, $C_{Hd} = \frac{1}{2} [\text{Var}^*(h_1) - \text{Var}^*(h_0) - V_d]$, $V_H = \text{Var}^*(h_0)$. Three waves; algebraically identical to the covariance-only solve (Appendix A, the denoising case).

Figure 15: Illustration of the denoising case



The general cases are Cases 3 and 4 in the main text. Case 3's noise layer (25) nests the denoising case at $\rho = 0$, Case 2 at constant v_a , and the random walk at $\rho = 1$; adding Case 4's spike, the family also contains Proposition D.1 below as its $\rho \rightarrow 1$ member, so the main text's class now covers every noise model used in this note. In practice we estimate that family plus a diagonal spike: minimum distance on (54) with the layers (25) + spike, with the main-text cases as the reading guide and the restrictions table below as the test menu.

The persistent-shocks case in full

Proposition D.1 (Random walk plus measurement noise). *Under A1-A2, let $u_{i,a} = w_{i,a} + \varepsilon_{i,a}$ with $w_{i,a} = w_{i,a-1} + \eta_{i,a}$, ε i.i.d. over ages, all innovation variances free to vary with age, and shocks mutually uncorrelated. The level covariances then satisfy, for $s < t$,*

$$\text{Cov}(h_s, h_t) = V_H + \text{Var}(w_0) + (s+t)C_{Hd} + stV_d + \sum_{\ell=1}^s \sigma_{\eta,\ell}^2 \quad (\text{the sum is empty for } s=0), \quad (56)$$

so fixing the earlier wave s and varying the later one leaves the noise terms untouched. With four waves:

$$C_{Hd} = \text{Cov}(h_0, h_2) - \text{Cov}(h_0, h_1) \quad (\text{off-diagonal read}), \quad (57)$$

$$V_d = [\text{Cov}(h_1, h_3) - \text{Cov}(h_1, h_2)] - C_{Hd}, \quad (58)$$

$$\sigma_{\eta,s}^2 = [\text{Cov}(h_s, h_t) - \text{Cov}(h_{s-1}, h_t)] - C_{Hd} - tV_d \quad (\text{any } t > s), \quad (59)$$

$$\sigma_{\varepsilon,a}^2 = \text{Var}(h_a) - \text{Cov}(h_a, h_t) + (t-a)(C_{Hd} + aV_d) \quad (\text{any } t > a). \quad (60)$$

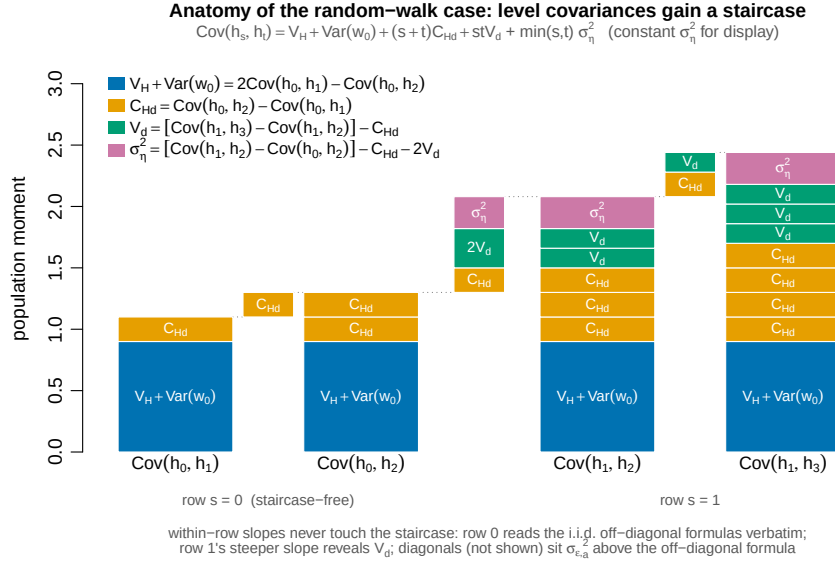
Levels identify V_H only in the bundle $V_H + \text{Var}(w_0) = 2\text{Cov}(h_0, h_1) - \text{Cov}(h_0, h_2)$, but that doesn't matter for the returns to prevention/treatment.

- Reading: the staircase makes the noise variance non-stationary, so the variance profile is unusable here and every read is off-diagonal. Row 0 keeps the denoising-case formulas because

its noise sum is empty; moving along a row adds $C_{Hd} + rV_d$ per wave; crossing to the next row additionally picks up one permanent-shock variance; the diagonal sits $\sigma_{\varepsilon,a}^2$ above the off-diagonal formula (Figure 16). Distance from the diagonal never kills the staircase - direction does: row reads are staircase-immune.

- Equivalent difference combinations: $\text{Cov}(\Delta h_t, \Delta h_{t+k}) = V_d$ and $\text{Cov}(h_0, \Delta h_k) = C_{Hd}$ for $k \geq 2$; at lag one, $\text{Cov}(\Delta h_t, \Delta h_{t+1}) = V_d - \sigma_{\varepsilon,t}^2$ (the mean-reversion dip).
- C_{Hd} needs 3 waves, V_d needs 4. Everything else is identified except at the panel's edge (the last wave's $\sigma_\eta^2/\sigma_\varepsilon^2$ split is identified only in sum).

Figure 16: Illustration of Proposition D.1



Proof of Proposition D.1. For $s < t$, $\text{Cov}(u_s, u_t) = \text{Cov}(w_s + \varepsilon_s, w_t + \varepsilon_t) = \text{Var}(w_s) = \text{Var}(w_0) + \sum_{\ell \leq s} \sigma_{\eta,\ell}^2$ (the walk is shared up to the earlier wave; the ε 's share nothing), which added to (5) gives the stated level covariances. The reads follow by differencing: along a row, $t \mapsto t+1$ at fixed s adds $C_{Hd} + rV_d$; across rows, $s \mapsto s+1$ at fixed t adds $C_{Hd} + tV_d + \sigma_{\eta,s+1}^2$; and $\text{Var}(h_a) - \text{Cov}(h_a, h_t) = \sigma_{\varepsilon,a}^2 - (t-a)(C_{Hd} + aV_d)$ gives the diagonal excess. Row 0 shows only the bundle: $\text{Cov}(h_0, h_t) = (V_H + \text{Var}(w_0)) + tC_{Hd}$. For the difference combinations: $\Delta h_t = d + \eta_t + \varepsilon_t - \varepsilon_{t-1}$; for $k \geq 2$ the noise terms share no common shock, so $\text{Cov}(\Delta h_t, \Delta h_{t+k}) = V_d$ and $\text{Cov}(h_0, \Delta h_k) = C_{Hd}$; at $k = 1$ the shared ε_t contributes $-\sigma_{\varepsilon,t}^2$. $\square$

The cases as restrictions

Each case is (53)/(54) with layers switched off; each added layer costs waves; each switched-off layer is a testable restriction.

	Noise restriction	Fingerprint	Test	Waves
$C_{Hd}=0$ remark	i.i.d., stationary	flat row 0	$C_0(1) = C_0(2)$	2
Case 1	i.i.d., stationary	quadratic diagonal	quadratic profile ($T>3$)	3
Case 2	AR(1), stationary	geometric fade off-diagonal	constant fade ratio	3
denoising	independent, free v_a	spike only	rows linear in t	3
Case 3	AR(1), free v_a	fade $\times$ row amplitude	constant S_k ratio	4
Case 4	AR(1) + spike	fade off-diagonal, ridge on	interior cells	4
Prop. D.1	random walk + i.i.d.	staircase + spike	flat D_k from $k \geq 2$	4

Waves = minimum for just-identification; running a row's test needs at least the next wave (the remark row's needs a third; the D_k ladder a fifth).

Limits

- *Two splits inside the equation.* V_H vs. ω^2 : both live only in the level B - for levels, a permanent stochastic position is a type intercept, which is the framework's own message and the $\rho \rightarrow 1$ limit of Section 3.2 (neither part is needed for the returns). And at the panel's edge, $G(T-1)$ appears only in the last diagonal cell, so the final wave's split is identified only in sum.
- *Persistence outside the class.* A γ_u outside the family (25) - for example Case 3 plus an i.i.d. spike, or MA(q) components - loses the clean reads. Identification is then by minimum distance on (54). A rising lag- ≥ 2 ladder D_k under the fitted class is the sign that the class is too small.
- *Distributional shape.* The surface sees only second moments of (H_i, d_i) ; the number and location of types, multimodality, and tail behaviour are invisible - clustering's job, and exactly what the targeting appendix needs (Appendix B).
- *Selection.* With health-related attrition or mortality the observed surface is conditional on survival, and the single-equation story needs reweighting or an explicit survival margin. Flagged, not solved.

E A first empirical pass at the local covariance (parked)

This appendix records a first attempt at estimating the local level-change covariance $C_{Hd}(a)$ of Section 3.4 from the UKHLS covariance rows, so that the results can be picked up again when the full empirical section is built. The main text deliberately stops at illustration; nothing before this appendix uses these estimates. The estimator library and its simulation gates (`emp_26_row_estimator_fns.R`, `emp_26_local_rows.R`, `check_row_estimator.R`) live with the empirical companion repo.

What was estimated

- For each base age s , the row $\text{Cov}(h_s, h_{s+k})$, $k = 0, \dots, 9$, was fitted per anchor; the slope of the deterministic part is $C_{Hd}(s)$.
- The fits differ only in how the noise is treated, and each was gated on simulated data before touching the data.
- Over a nine-year row a slow fade, a curve and a straight line partially mimic one another, so no single treatment is unbiased under every noise process; a suite is used and its spread is reported as model uncertainty.

- The suite: a *bundle* fit (one-year spike + quadratic, no decay column; exact when the persistent noise is near-permanent - the corner the data itself selects, and the $\rho \rightarrow 1$ bundle limit of Section 3.2); a per-anchor quadratic-plus-decay fit (exact under pure AR(1) noise; biased late under spike mixtures); a line-plus-decay fit (reads the row-average local covariance; the early side of the bracket); the four-cell just-identified read of (27); and a common- ρ joint fit (recorded, but degenerate here: the data pushes ρ to the top of the grid, where the decay column is collinear with the line).
- Person-level bootstrap (300 replicates) supplies the confidence bands; row lengths $K = 6, 9, 12$ as sensitivity.

What the data said

- On the primary measure, $C_{Hd}(a)$ sits at or just below zero through ages 20-50 and turns decisively negative from the late 50s, reaching -0.002 to -0.004 per year by 70-80 (Figure 17).
- The same shape appears under every treatment in the suite and on every measure (Figure 18); the flattest is the SF-6D, as elsewhere.
- A single “crossing age” is not a robust object: the young-age sign straddles zero, so the formal zero-crossing moves between about 55 and 71 with the noise treatment. The robust statements are the break and the late-life magnitude.
- The noise anatomy is a finding in itself: a large one-year spike plus a near-permanent slow layer. Profiling the fade speed runs to the top of the grid - the bundle limit of Section 3.2 operating in the data - which is why the bundle fit is the primary. This structure is Case 4 of Section 3.3 at its $\rho \rightarrow 1$ corner: what Case 4 identifies cleanly with a fourth wave when ρ is away from one, the suite here handles at the corner by folding the carried part into the level.
- The spike *level* rises with age (roughly 0.005 to 0.009 on the primary measure over ages 20-80); it is the spike’s *share* of the variance that falls.
- Corroboration from changes: the D_k plateau by age band is most negative at ages 60-70 on the PCS measures - the change-based signature of the same compression.

Caveats carried forward

- Every row conditions on being observed k years later, so these are survivors’ covariances; the selection, floor and cohort-pooling treatments belong to the empirical section and have not been applied here.
- With the young-age covariance at zero rather than positive, compounding is weak at best on these measures; the strong empirical fact is the compression after 60.

Figure 17: The local covariance $C_{Hd}(a)$ on the primary measure: bundle fit with 95% bootstrap band, and the spike level by age.

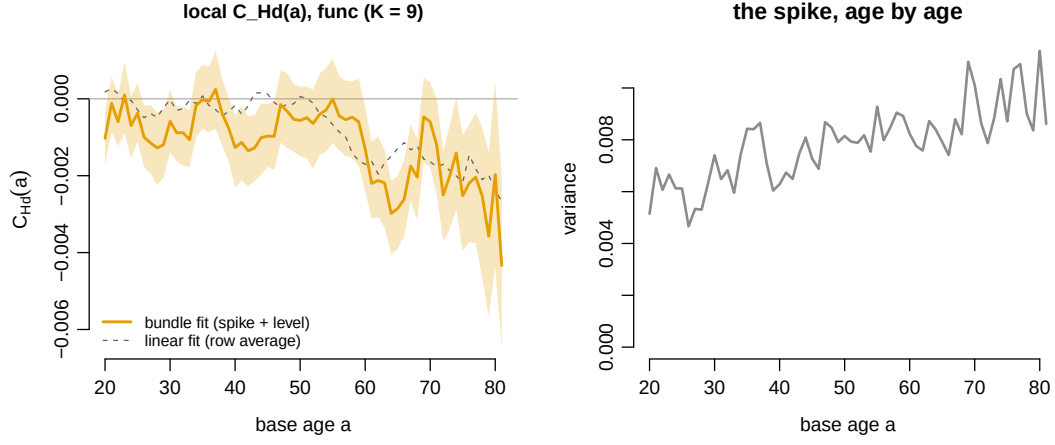


Figure 18: The same profile across the six measures ($K = 9$).

